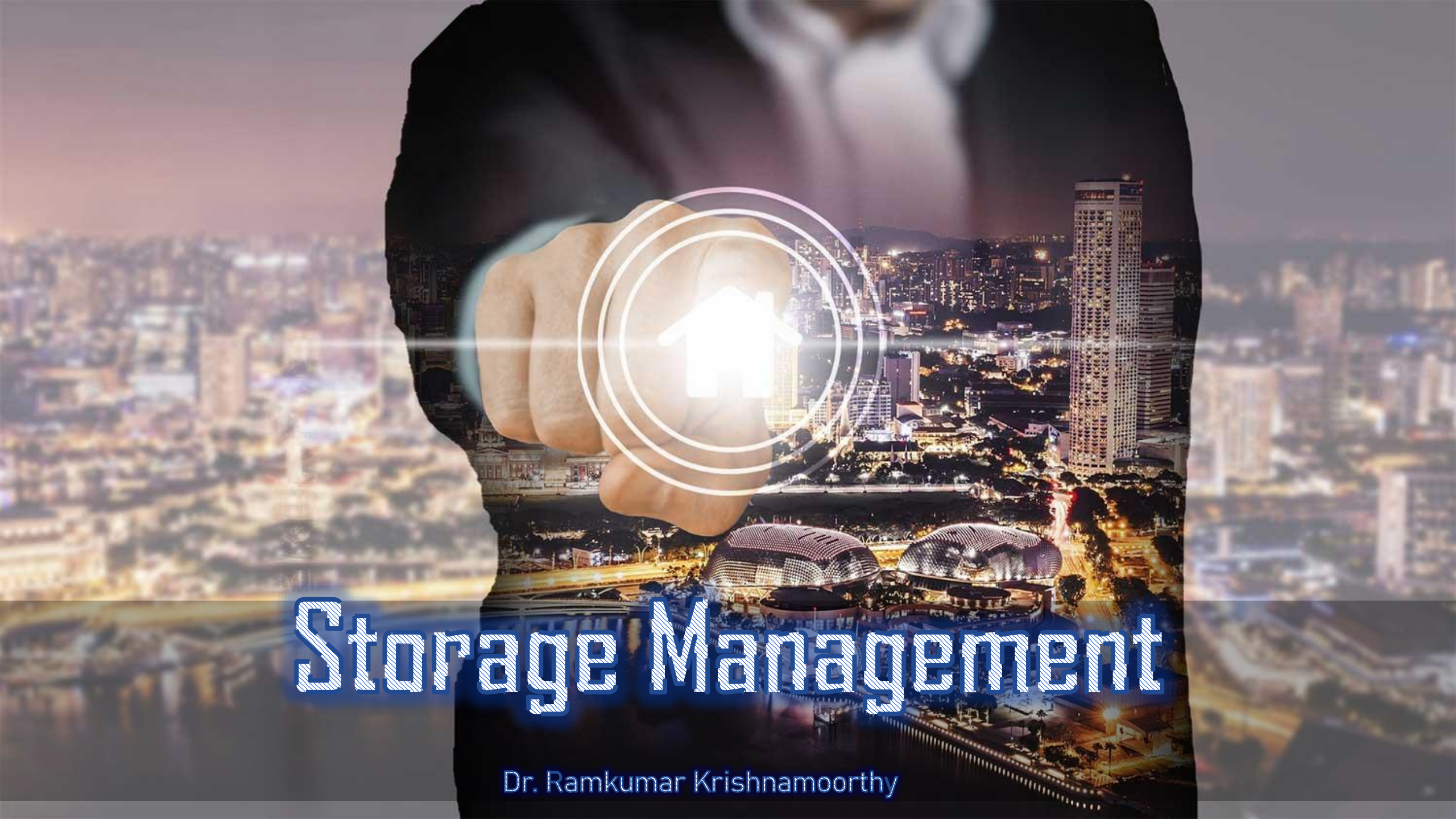


A hand in a dark suit jacket points towards a glowing white house icon. The icon is surrounded by three concentric white circles. The background is a blurred night cityscape with lights from buildings and streets. The overall image has a dark, moody aesthetic with a focus on the hand and the house icon.

Storage Management

Dr. Ramkumar Krishnamoorthy

Contents

- Storage System Environment: Components of a Storage System Environment – Host –Connectivity – Storage, Disk Drive Components –Platter – Spindle
- Read/Write Heads - Actuator Arm Assembly - Controller - Physical Disk Structure -SSD & HDD- Types of hard drive connections
- Blade server and Rack server -Zoned Bit Recording - Logical Block Addressing, Disk Drive Performance -1 Disk Service Time
- Fundamental Laws Governing Disk Performance , Logical Components of the Host - Operating System -Device Driver -Volume Manager
- File System – Application , Application Requirements and Disk Performance.

Components of a Storage System Environment

There are three main components in a storage system environment:

- **Host:** The host system is used for the interaction between the OS and application that has requested the data.
- **Connectivity:** It provides the connection and helps in carrying out the data and the read or write commands between the storage mediums and the host.
- **Storage:** The data gets stored into the storage mediums or devices.

Components of a Storage System Environment

Host

It is one of the main components in a storage system environment:

- Applications run on hosts
- Hosts can range from simple laptops to complex server clusters
- Physical components of a host:

1. CPU

2. Storage

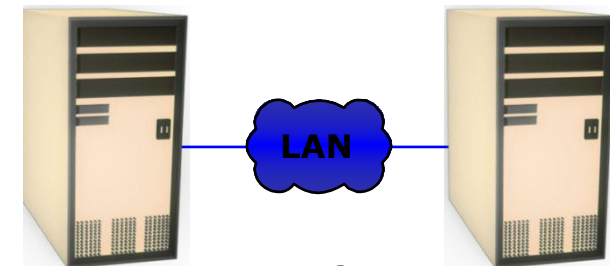
- Disk device and internal memory.

3. I/O device

- Host to host communications
 - Network Interface Card (NIC).
- Host to storage device communications
 - Host Bus Adapter (HBA).



Server



Group of Servers

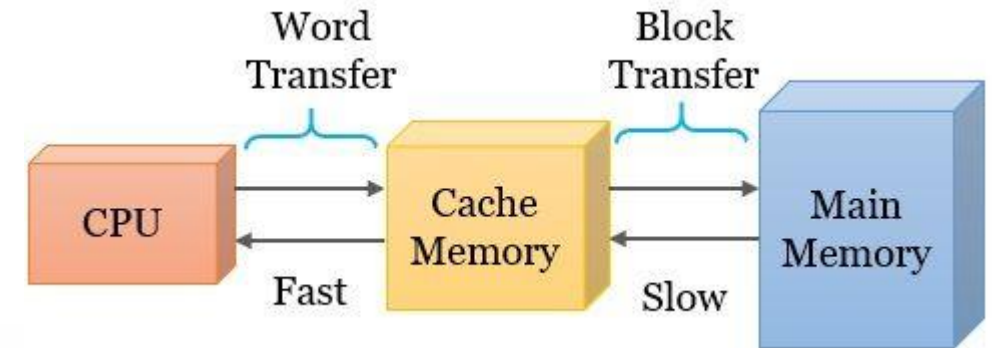
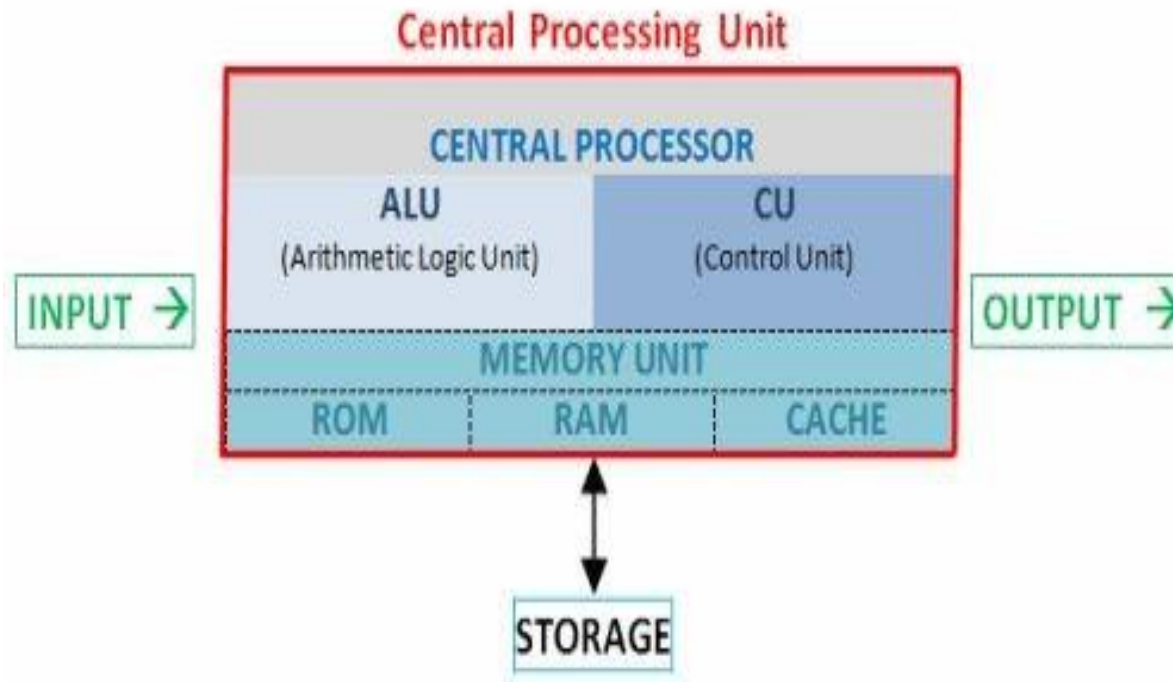


Components of a Storage System Environment

CPU

The CPU consists of four main components:

- Arithmetic Logic Unit (ALU).
- Control Unit.
- Register.
- Cache.



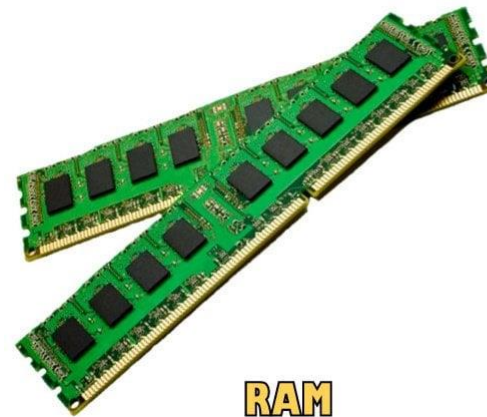
Register

Components of a Storage System Environment

Storage

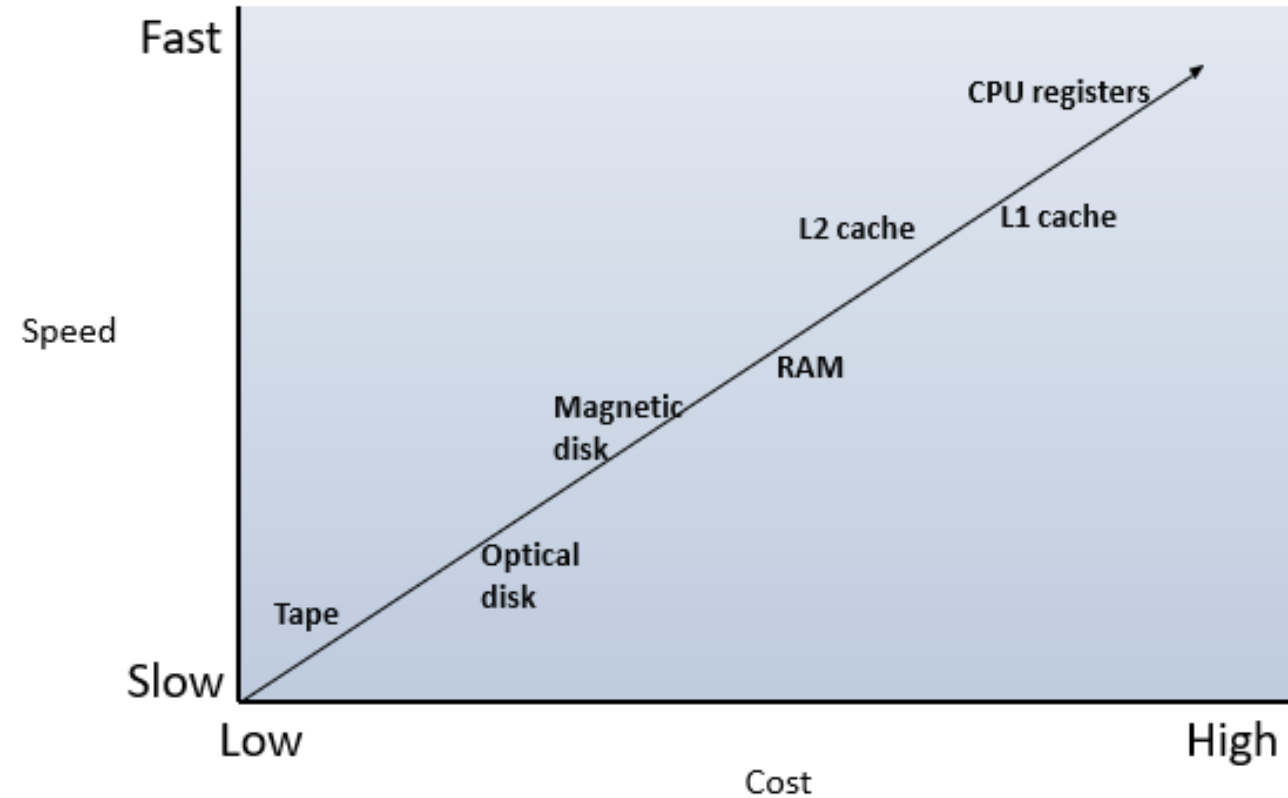
Memory and storage media are used to store data, either persistently or temporarily. Generally, there are two types of memory on a host:

- Random Access Memory (RAM) – Static RAM (holds permanently) and Dynamic RAM (Refreshes in periodically)
- Read-Only Memory (ROM) – PROM, EPROM (Uses UV waves), Electronically EPROM (Electric Signals)
- Storage devices are less expensive than semiconductor memory. Examples of storage devices are as follows:
 - Hard disk (magnetic).
 - CD-ROM or DVD-ROM (optical).
 - Floppy disk (magnetic).
 - Tape drive (magnetic).



Components of a Storage System Environment

Components of Storage



Components of a Storage System Environment

I/O Devices

Human interface

- Keyboard
- Mouse
- Monitor

Computer-computer interface

- Network Interface Card (NIC)

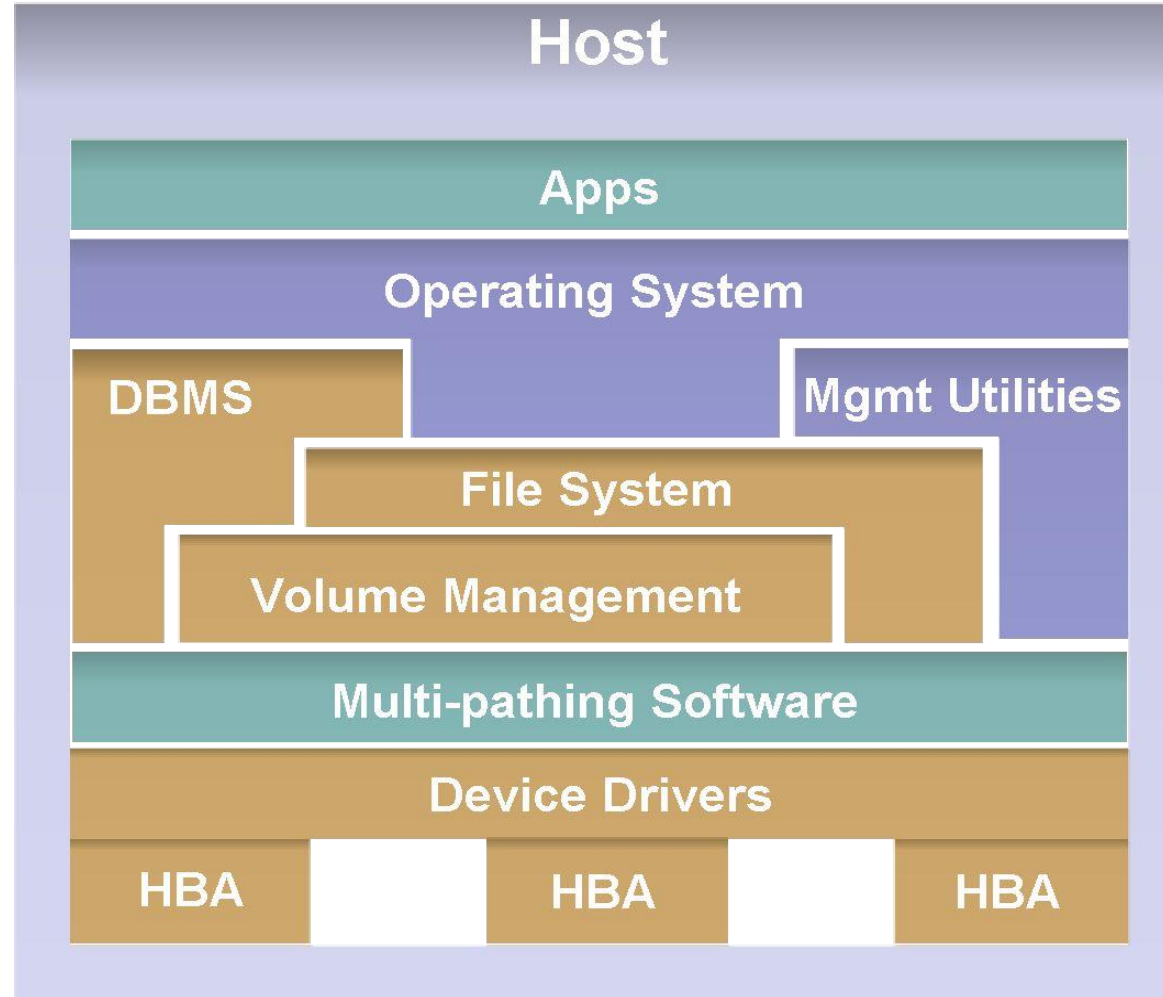
Computer-peripheral interface

- USB (Universal Serial Bus) port
- Host Bus Adapter (HBA)



Components of a Storage System Environment

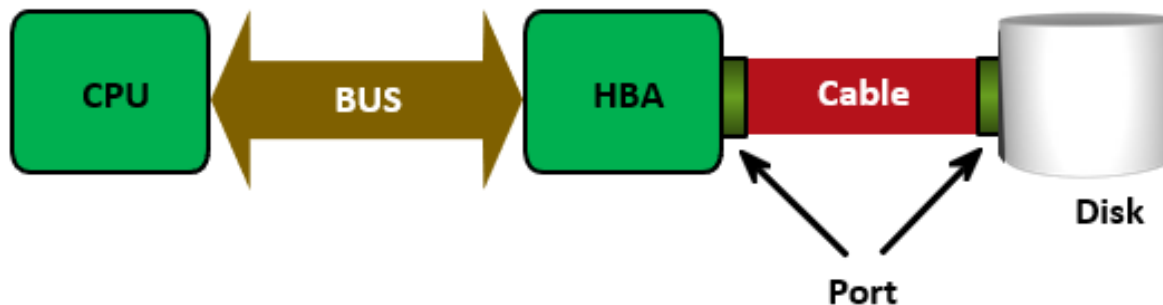
Logical Components of Host



Connectivity

Connectivity

- Provides interconnection between hosts or between a host and any storage devices
- The physical components are the hardware elements that connect the host to storage, and the logical components of connectivity are the protocols used for communication between the host and storage.
- Physical Components of Connectivity are: Bus, Port and Cable



Connectivity

Physical components of connectivity bus, port and cables

- The bus is the collection of paths that facilitate data transmission from one part of a computer to another, such as from the CPU to the memory.
- The port is a specialised outlet that enables connectivity between the host and external devices.
- Cables connect hosts to internal or external devices using copper or fibre optic media.

Buses, as conduits of data transfer on the computer system, can be classified as follows:

- **System bus:** The bus that carries data from the processor to memory.
- **Local or I/O bus:** A high-speed pathway that connects directly to the processor and carries data between the peripheral devices, such as storage devices and the processor.



Connectivity

Bus Technology

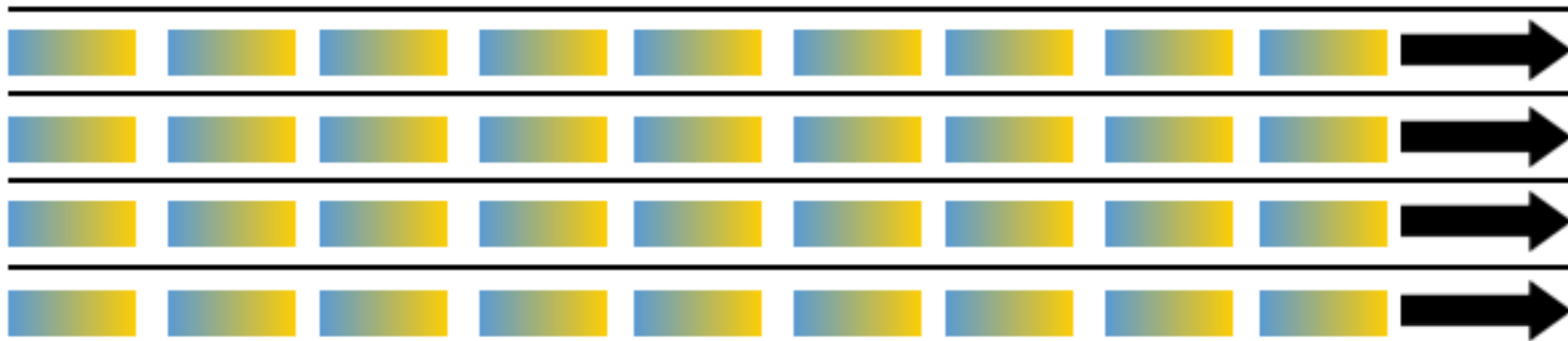


Serial

Serial Bi-directional



Parallel



Logical Components of Connectivity

Logical Components of Connectivity

- The popular interface protocol used for the local bus to connect to a peripheral device is a Peripheral Component Interconnect (PCI).
- The interface protocols that connect to disk systems are **Integrated Device Electronics/Advanced Technology Attachment (IDE/ATA)** and **Small Computer System Interface (SCSI)**.
- PCI is a specification that standardises how PCI expansion cards, such as network cards or modems, exchange information with the CPU. PCI provides the interconnection between the CPU and attached devices.
- IDE/ATA is the most popular interface protocol used on modern disks. This protocol offers excellent performance at relatively low cost.
- SCSI has emerged as a preferred protocol in high-end computers. This interface is far less commonly used than IDE/ATA on personal computers due to its higher cost.

Storage

Storage

- The storage device is the most important component in the storage system environment.
- A storage device uses magnetic or solid state media.
- Disks, tapes, and diskettes use magnetic media.
- CD-ROM is an example of a storage device that uses optical media, and a removable flash memory card is an example of solid-state media.

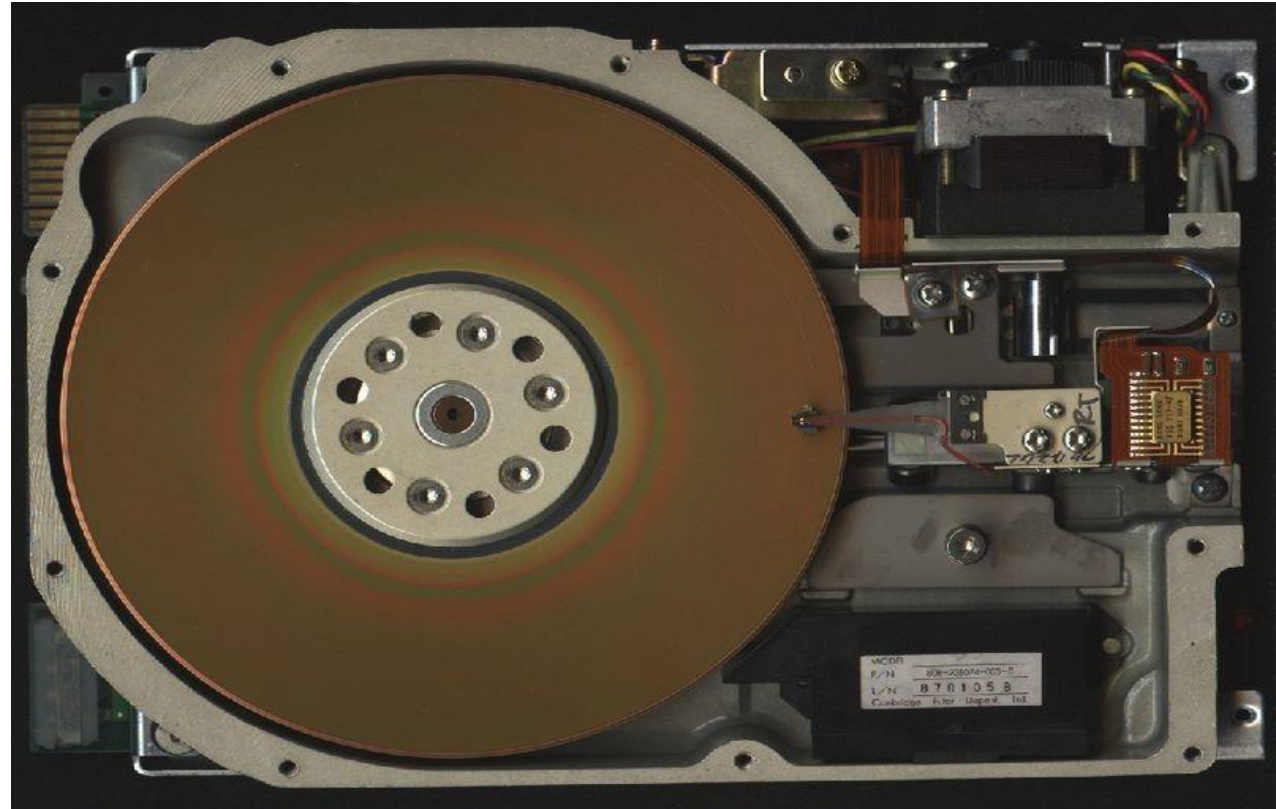
Components of Storage

Magnetic Media

- Magnetic storage is an example of non-volatile memory, which means when the storage device is not powered, then the data does not get lost and is retained.
- Magnetic storage uses various patterns of magnetisation in a magnetisable material to store the data.
- Magnetic storages are relatively cheap and are widely used for data storage.
- Examples of devices that use the magnetic media are tapes, disks, and diskettes.

Components of Storage

Magnetic Disk Drive(Hard disk drive)



Components of Storage

Optical Media

- Optical media are the kind of storage media that contain the content in a digital format and the read/write feature can be performed on them using a laser.
- The examples of optical media include CD-ROMs, DVD-ROMs, and all the variations of the two formats, as well as optical jukeboxes.
- Optical media can last up to seven times as long as any traditional storage media.



Components of Storage

Solid state media

- It is made up of silicon microchips. It is non-volatile storage and stores data electronically instead of storing it magnetically.
- It contains no mechanical parts, which allows the transfer of data, to and from the storage medium, to take place at a very higher speed.
- Provides data integrity and endurance as any other electronic device.
- An example of a device that uses the solid-state media is a removable flash memory.



Components of Storage

Tapes

- Tapes are a popular storage media used for backup because of their relatively low cost. In the past, data Centres hosted a large number of tape drives and processed several thousand reels of tape. However, the tape has the following limitations:
 - Data is stored on the tape linearly along the length of the tape. As a result, **random data access is slow and time-consuming.**
 - In a shared computing environment, **data stored on tape cannot be accessed by multiple applications simultaneously**, restricting its use to one application at a time.
 - On a tape drive, the read/write head touches the tape surface, so the tape degrades or wears out after repeated use.
 - The storage and retrieval requirements of data from tape and the overhead associated with managing tape media are significant.

Components of Storage

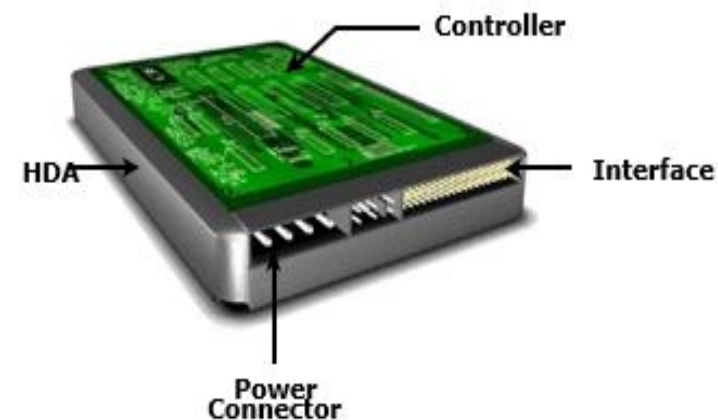
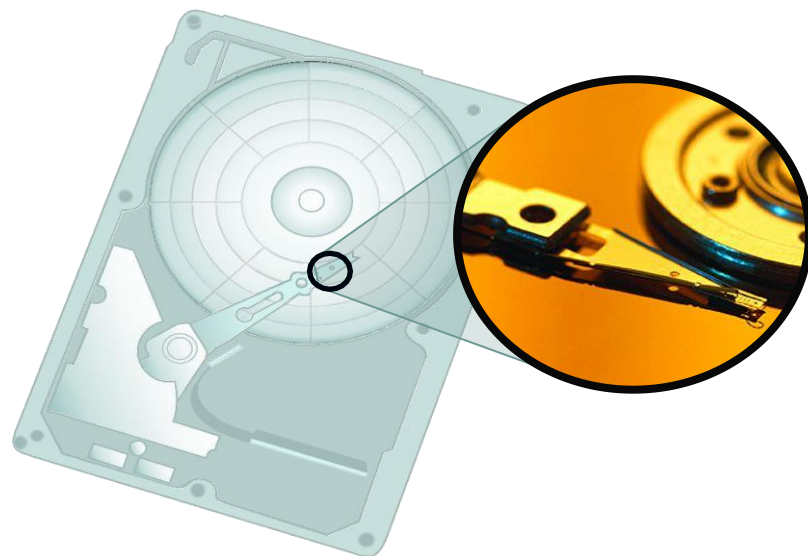
Magnetic Tapes



Disk Drivers

Disk Drive

A **disk drive** is one of the most popular, randomly accessible, and rewritable storage medium used in the modern computers. **It is a physical drive in a computer which is capable of holding and retrieving information.** The disk drives are used for storing and accessing the data for a performance-intensive, online application.

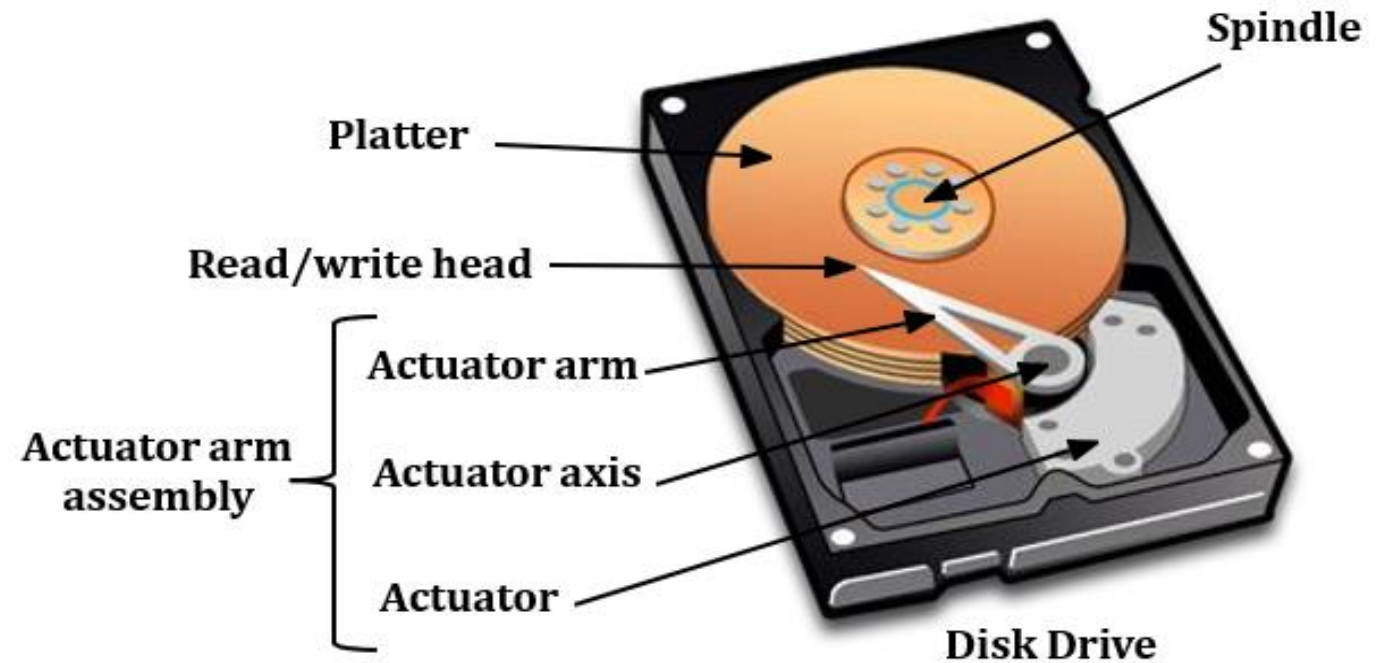


Disk Drivers Components

Components of a Disk Drive

The key components of a disk drive are:

- Platter
- Spindle
- Read/write head
- Actuator arm assembly
- Controller



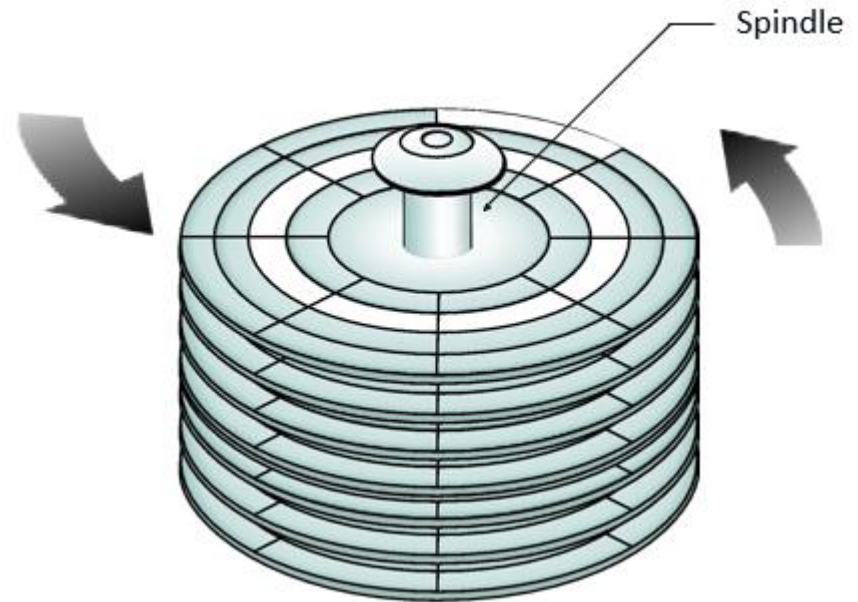
Disk Drivers Components

Platters



A typical HDD consists of one or more flat circular disks called platters. The data is recorded on these platters in binary codes (0s and 1s)

Spindle

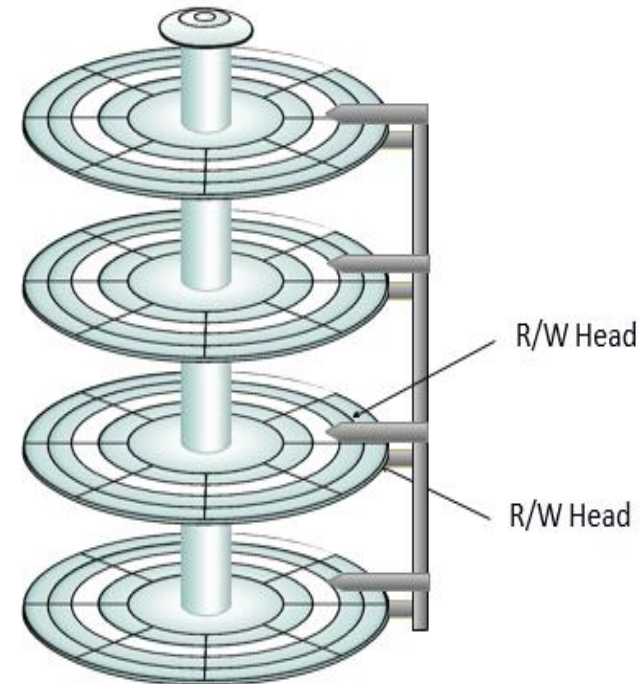


A spindle connects all the platters and is connected to a motor. The motor of the spindle rotates with a constant speed

Disk Drivers Components

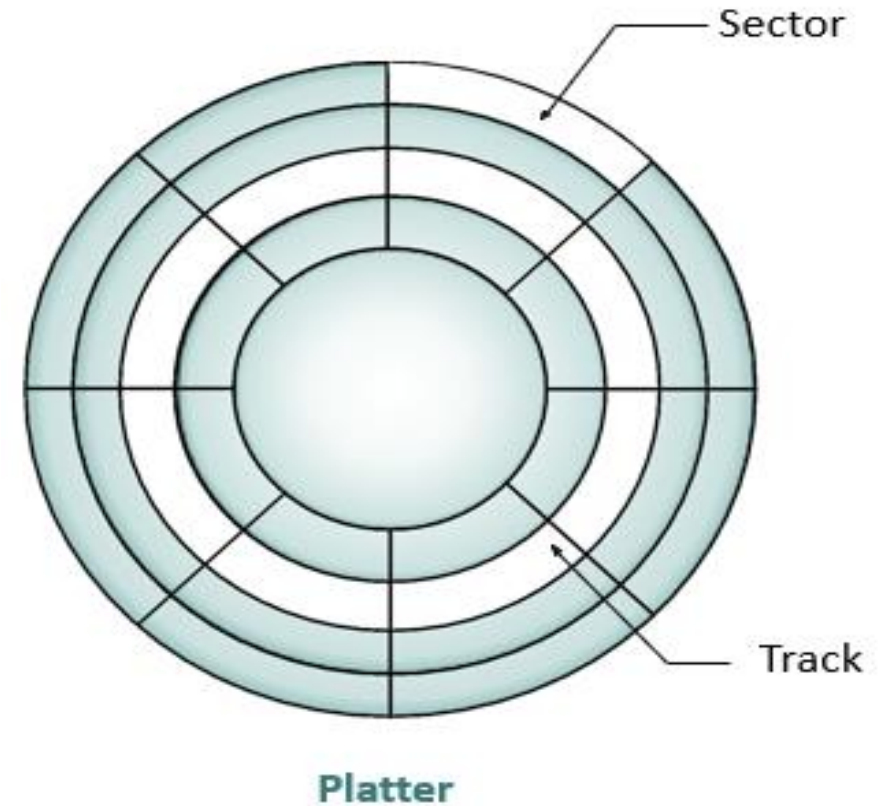
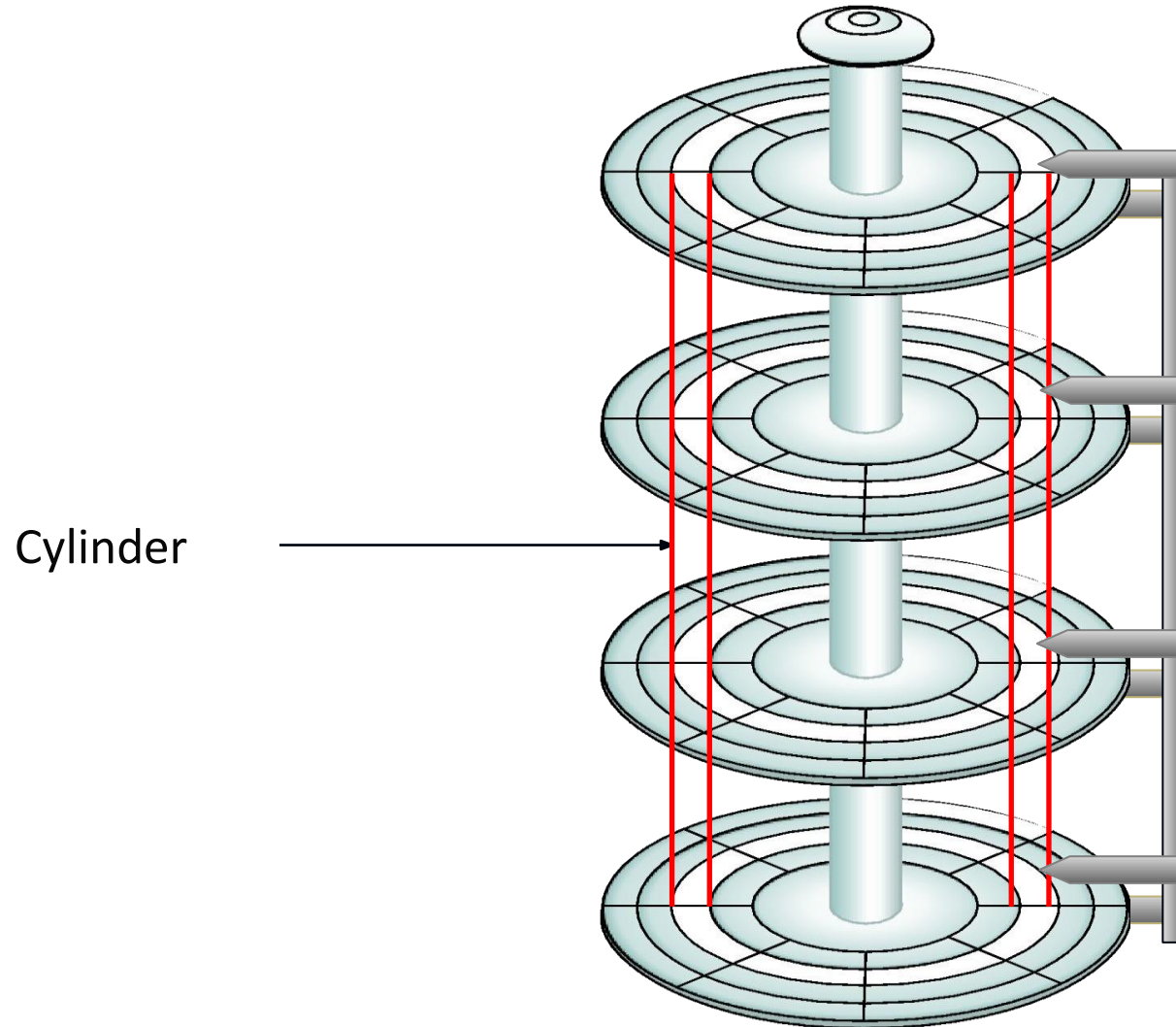
Read/Write Heads

- Data is read and written by read/write heads, or R/W heads.
- Most drives have two R/W heads per platter, one for each surface of the platter.
 - When reading data, they detect magnetic polarisation on the platter surface.
 - When writing data, they change the magnetic polarisation on the platter surface



Disk Drivers Components

Sectors and Tracks



Disk Drivers Components

Actuator Arm Assembly

- The portion of the device which tacks the read/write head of the defined path.
- Actuator Arm physically transfers the actuator's action to the mechanism that it is designed to operate.



Disk Drivers Components

Controllers

- It is an electrical component in a computer's hard disk that allows the processor to access, read, write, delete, and modify data to and from the hard disk



Bottom View of Disk Drive

Solid State Disk (SSD)

- SSD is a non-volatile storage device, which stands for Solid State Drive. SSD stores the data on flash memory chips and maintains the data in a permanent state, even when the power is off.
- SSDs do not have any moving parts. That's why they are called solid-state drives
- **Smaller Physical Size** – which could also be removable in the case of CompactFlash, SD Card, etc.
- **Lower Power Usage**
- **Higher Performance** – especially small block transfers used for transactional processing
- **Lower Absolute Cost** – if only a small amount of memory is needed



Features of Solid State Disk (SSD)

- **Velocity:** SSDs are quicker than HDDs because they use flash memory. This means user computer starts quickly, programs open quicker, and everything feels more responsive.
- **Reliability:** SSDs are more reliable because they don't have moving parts like HDDs. HDDs can get damaged if dropped or bumped, but SSDs can handle these better.
- **Durability:** SSDs are tough and can handle rough conditions like extreme temperatures and vibrations. They're appropriate for laptops, portable devices, and places where reliability and durability count.

Features of Solid State Disk (SSD)

- **Energy efficient:** SSDs use less power because they don't have spinning disks or moving components like HDDs. This means laptops and portable devices can last longer on a single battery charge.
- **Noiseless:** SSDs are silent because they have no moving parts that can make noise. This makes them great for locations where you need quiet things, like recording studios, offices, and home theaters.
- **Small size:** SSDs come in specific small sizes, like M.2 and mSATA. This indicates they absorb less space in user device compared to HDDs. That's why they're good for thin and lightweight devices like ultrabooks, tablets, and mini-computers.

Types of Solid State Disk (SSD)



SATA

Serial AT Attachment



PCIe

Probe-based confocal laser endomicroscopy



NVMe

Non-volatile memory express



M.2



U.2



External SSD

Hard Disk Drive (HDD)

- A computer hard disk drive (HDD) is a non-volatile data storage device. Non-volatile refers to storage devices that maintain stored data when turned off. All computers need a storage device, and HDDs are just one example of a type of storage device.
- HDDs are usually installed inside desktop computers, mobile devices, consumer electronics and enterprise storage arrays in data centers. They can store operating systems, software programs and other files using magnetic disks.

Hard Disk Drive (HDD) – Storage

Some of the most common storage drive capacities include the following:

- **16 GB, 32 GB and 64 GB.** This range is among the lowest for HDD storage space and is typically found in older and smaller devices.
- 120 GB and 256 GB. This range is generally considered an entry point for HDD devices such as laptops or computers.
- 500 GB, 1 TB and 2 TB. Around 500 GB and above of HDD storage is typically considered decent for an average user. Users can most likely store all their music, photos, videos and other files with this much space. Individuals with games that take up a lot of space should find 1 TB to 2 TB of HDD space suitable.
- More than 2 TB. Anything over 2 TB of HDD space is suitable for users who work with high-resolution files, who need to store or house a large amount of data, or who want to use that space for backup and redundancy.

Types of Hard Disk Drive (HDD)



Internal HDD and External HDD

Hard Disk Drive (HDD)

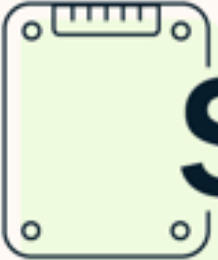
Advantages of an HDD:

- HDDs can store a large amount of data (this varies by the drive size)
- They are relatively cheap compared to other storage solutions

Disadvantages of an HDD:

- HDDs can be slow at retrieving larger files
- They consume more power
- The moving parts produce a lot of heat
- They are less durable, especially in portable devices

SSD vs HDD

	SSD vs HDD		
faster	✓	✗	slower
more expensive	✗	✓	cheaper
non-mechanical (flash)	✓	✗	mechanical (moving parts)
shock-resistant	✓	✗	fragile
best for storing operating systems, gaming apps, and frequently used files	i		best for storing extra data, such as movies, photos, and documents

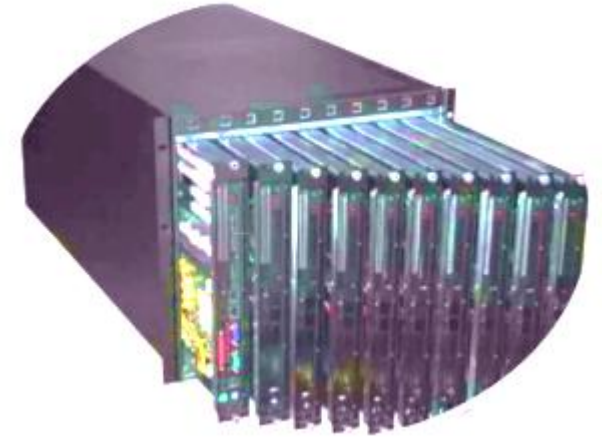
Blade Server

- A blade server, sometimes referred to as a high-density server, is a compact device containing a computer used to manage and distribute data in a collection of computers and systems, called a network. Its role is to act as a conduit (tube/channel) between computers, programs, applications and systems.

The benefits of using blade servers include:

- **Reduced power consumption.** Blade chassis provide power to many servers, reducing total consumption.
- **Increased processing power.** Blade servers allow for extremely high processing power in a minimal amount of physical space.
- **Easily serviceable.** Blade servers can be hot swappable, meaning if one server requires maintenance, it can easily be pulled and replaced.
- **Reduced cabling.** Blade servers require less cabling than rack servers.

Blade Server



In Each Blade Server

- A min 2 CPUs
- Memory
- Internal Storage
- Some Plugin Components



Rake Sever

- A rack server, or rack-mounted server, is any server that is built specifically to be mounted within a server rack.
- Rack servers are a general-purpose machine that can be configured to support a wide range of requirements. They are most commonly found in datacenter environments but can also be used in smaller computer.
- Generally, rack servers are 19 inches in width but can also come in 23 and 24-inch width.



Blade Server vs. Rack Server

Blade Servers	Comparison	Rack Servers
Best	Scalability	Good
Good	Cooling	Bad
Best	Resources	Bad
Large	Size of Business	Small and Medium
Bad (Parts cannot be Replaced Easily)	Flexibility	Good
Smaller than Rack Servers	Space	Small Horizontally and Vertically
Best (Hot Plugging, has Handles, Resources Centralization)	Management & Maintenance	Good



BLADE



RACK



Applications

Blade Server Applications

- Web page serving and caching.
- Streaming audio and video content.
- File sharing and serving.
- Terminal services.
- Email messaging.
- Databases hosting.
- Virtualization.

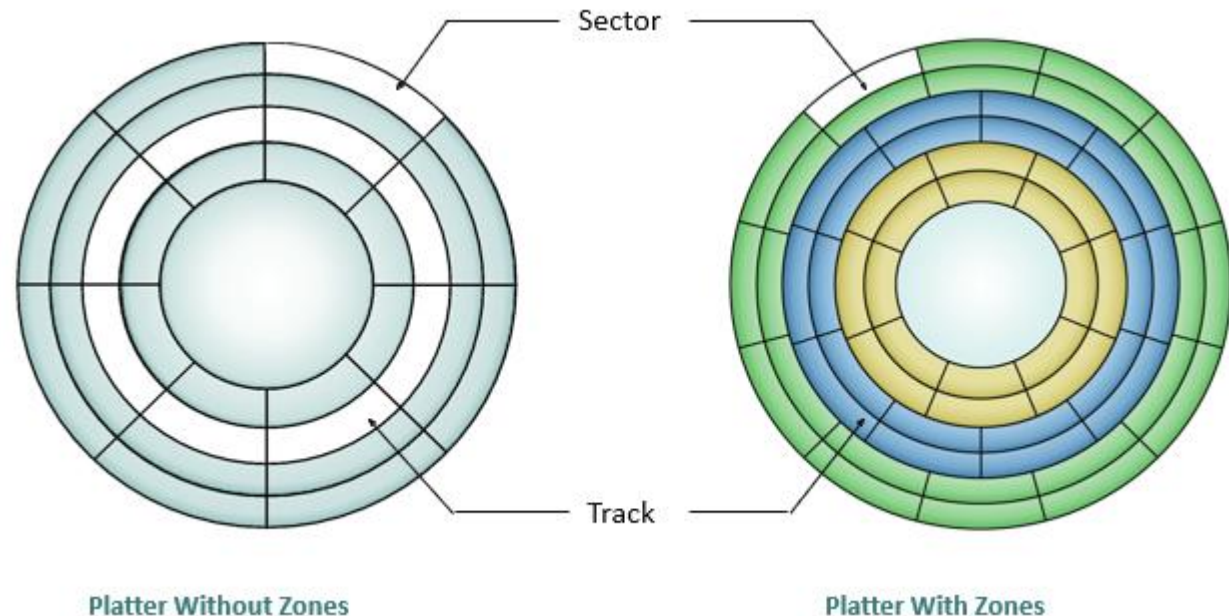
Rake Server Applications

- Military and Industrial Applications
- Embedded Computing
- Commercial warehouse
- Security Cameras

Zoned Bit Recording

Zoned-Bit Recording

- Since a platter is made up of concentric tracks, **the outer tracks can hold more data than the inner ones because they are physically longer than the inner tracks**. However, in older disk drives, the outer tracks had the same number of sectors as the inner tracks, which means that the data density was very low on the outer tracks. This was an inefficient use of the available space.
- **Zoned-bit recording** uses the disk more efficiently. It group tracks into zones that are based upon their distance from the Centre of the disk.



Logical Block Addressing

Logical Block Addressing

Earlier, drives used physical addresses made up of the Cylinder, Head, and the Sector number (CHS) to refer to specific locations on the disk. This meant that the host had to be aware of the geometry of each disk that was used.

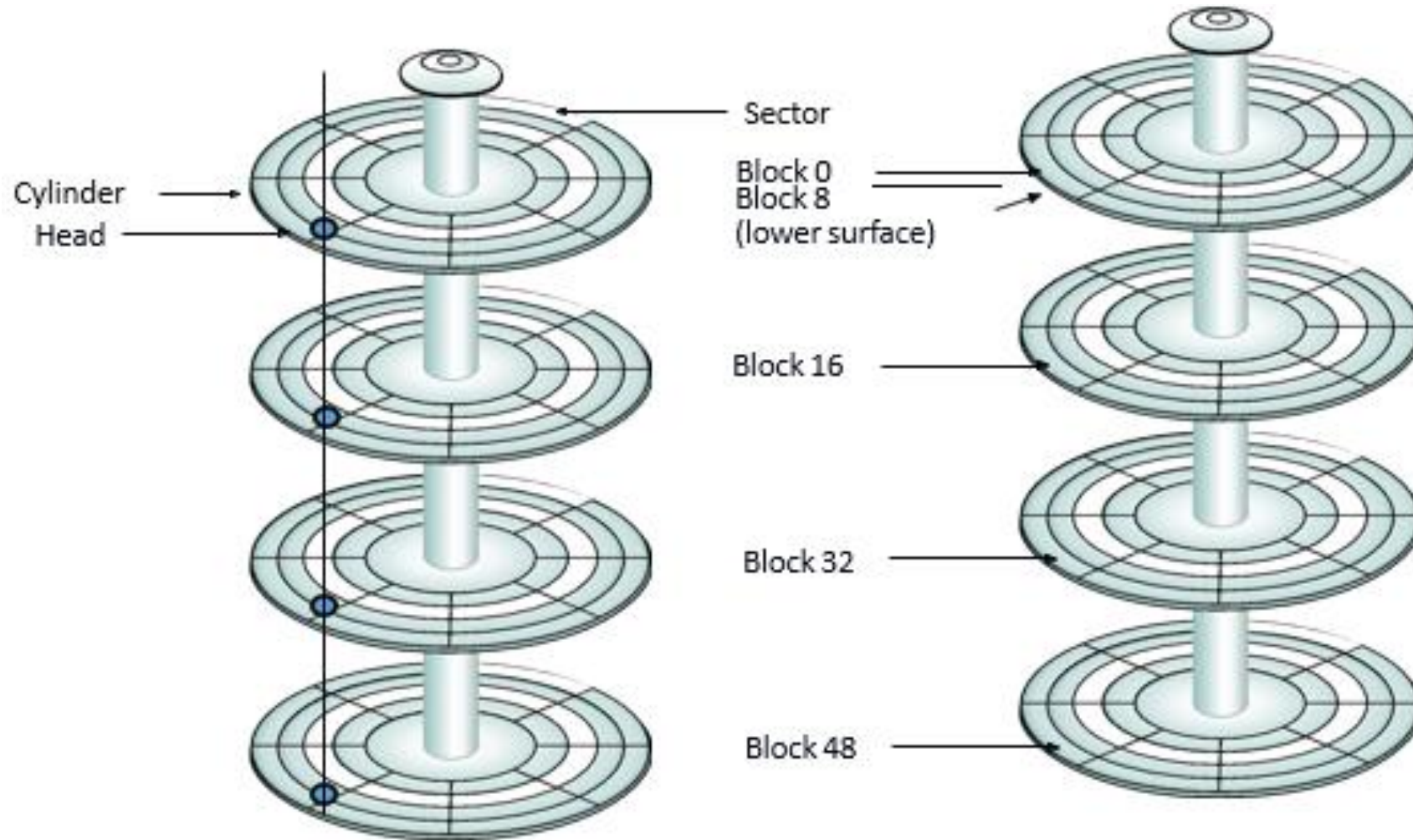
Logical Block Addressing (LBA) simplifies addressing by a **using a linear address for accessing physical blocks of data**. The disk controller performs the translation process from LBA to CHS address. The host only needs to know the size of the disk drive (how many blocks).

Logical blocks are mapped to physical sectors on a 1:1 basis

Block numbers start at 0 and increment by one until the last block is reached [E.g., 0, 1, 2, 3 ... (N-1)]

Block numbering starts at the beginning of a cylinder and continues until the end of that cylinder

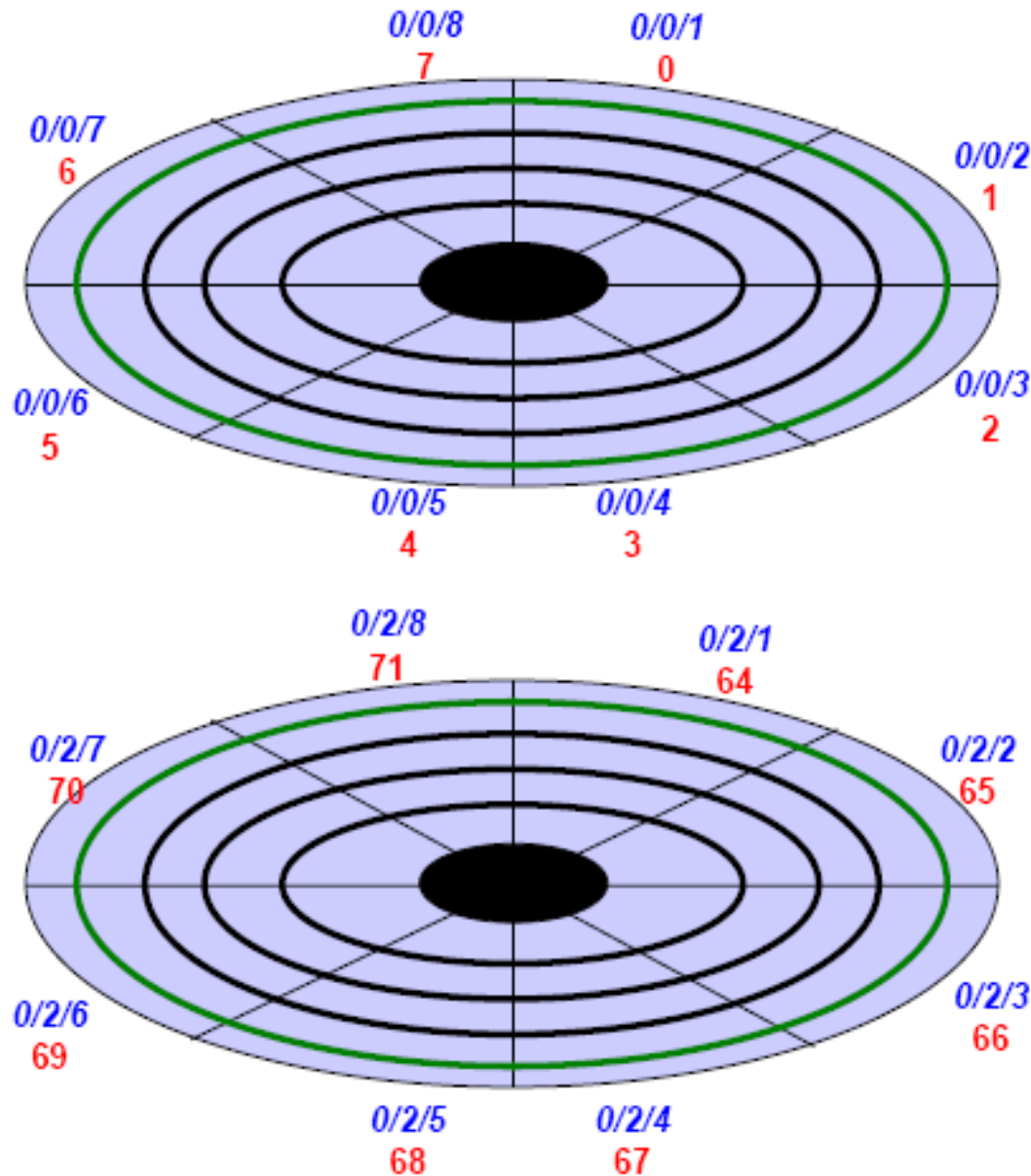
Logical Block Addressing



Physical Address = CHS

Logical Block Address = Block #

Logical Block Addressing



Simple example: 4 sides, 8 sectors/side, 4 tracks/side (cylinders) yields 128 total sectors...

The green track is cylinder 0. On the first side (side 0), sectors are numbered 0/0/1 thru 0/0/8 in CHS format, or 0 thru 7 in LBA.

On cylinder 1 of the first side, sectors are numbered 1/0/1 thru 1/0/8 (8 thru 15).

The last sector on side 0 is numbered 4/0/8 (31).

Second side (side 1) starts: 0/1/1 32
Second side ends: 4/1/8 63

Third side (side 2) starts: 0/2/1 64
Third side ends: 4/2/8 95

Fourth side (side 3) starts: 0/3/1 96
Fourth side ends: 4/3/8 127

Logical Block Addressing

LBA (Logical Block Addressing)

- This is the simpler, modern system. Instead of thinking about physical locations, LBA just numbers every single sector on the entire hard drive, starting from 0.
- Look at the numbers in red on the diagram:
- The sectors on **Side 0**, Cylinder 0 are numbered from **0** to **7**.
- The sectors on **Side 1**, Cylinder 0 are numbered from **8** to **15**.
- This continues all the way to the last sector, which is **127**.
- The computer uses a formula to convert an LBA number (like 32) back to its CHS address (0/1/1) to physically find the data.

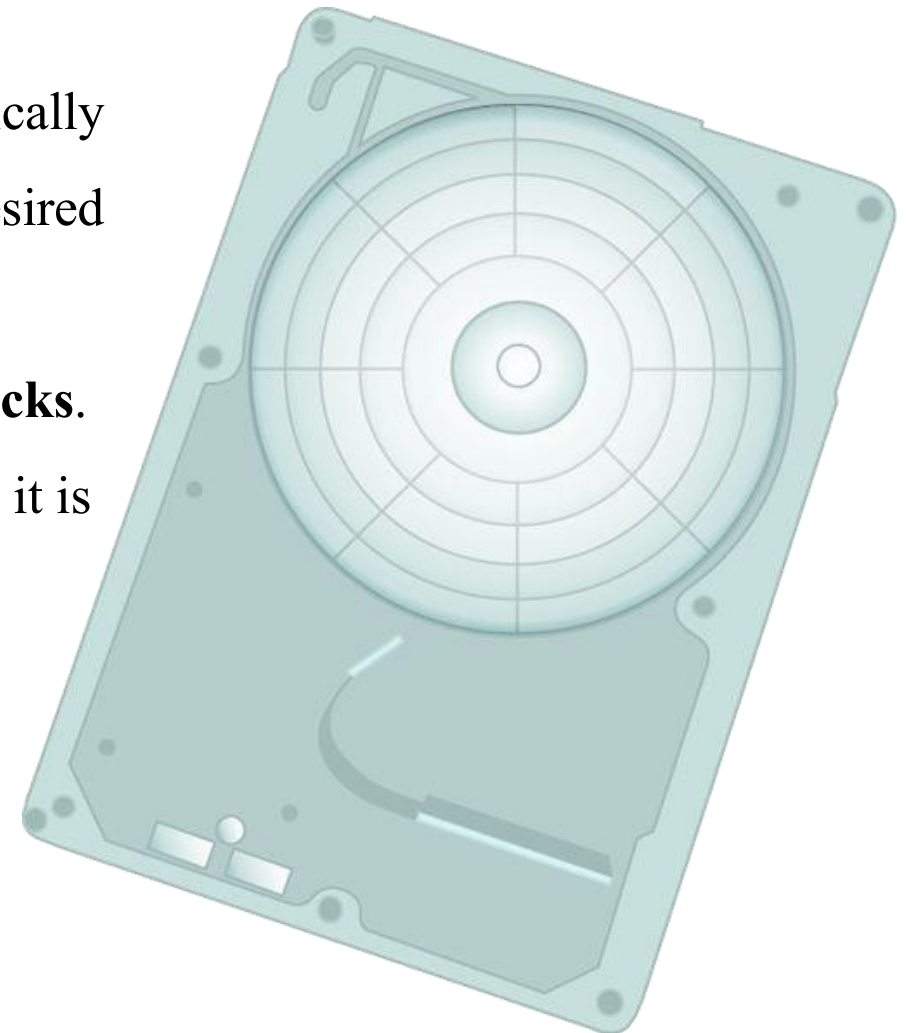
Example for the previous Picture

- The image provides a summary example:
- **4 sides:** (Sides 0, 1, 2, 3)
- **8 sectors/side:** (Sectors 1-8 for each side)
- **4 tracks/side (cylinders):** (Tracks 0, 1, 2, 3)
- The total number of sectors is calculated as: $4 \text{ sides} \times 8 \text{ sectors/track} \times 4 \text{ tracks} = 128 \text{ total sectors}$
- The image shows you how the LBA numbers map to the physical locations:
- The first 8 sectors (0-7) are on **Side 0**, Track 0.
- The next 8 sectors (8-15) are on **Side 1**, Track 0.
- The next 8 sectors (16-23) are on **Side 2**, Track 0.
- And so on, wrapping around the cylinders and sides until it reaches the final sector, LBA **127** (which is at the physical address **4/3/8**)

Disk Drive Performance – 1

Seek Time

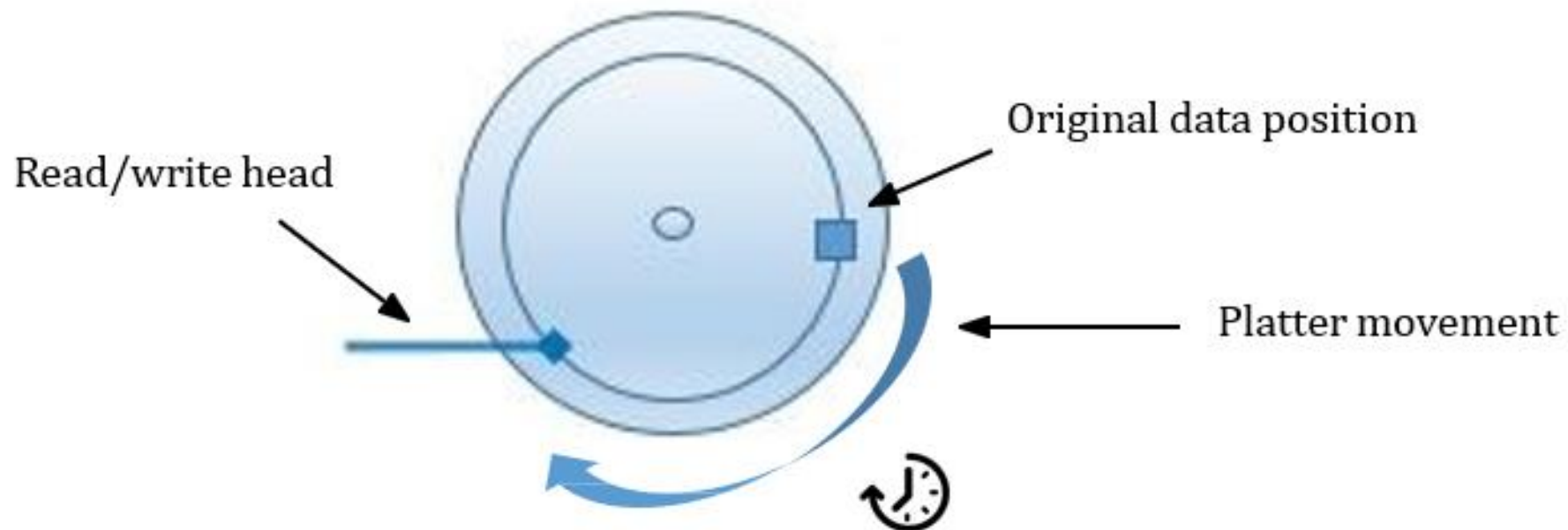
- The mechanical time it takes for the read/write head to physically move from its current position to the correct track where the desired data is stored
- Seek time is the time for **read/write heads to move between tracks.**
- Seek time is an important performance metric for HDDs because it is often the most significant factor in overall data access speed



Disk Drive Performance – 1

Rotational Latency

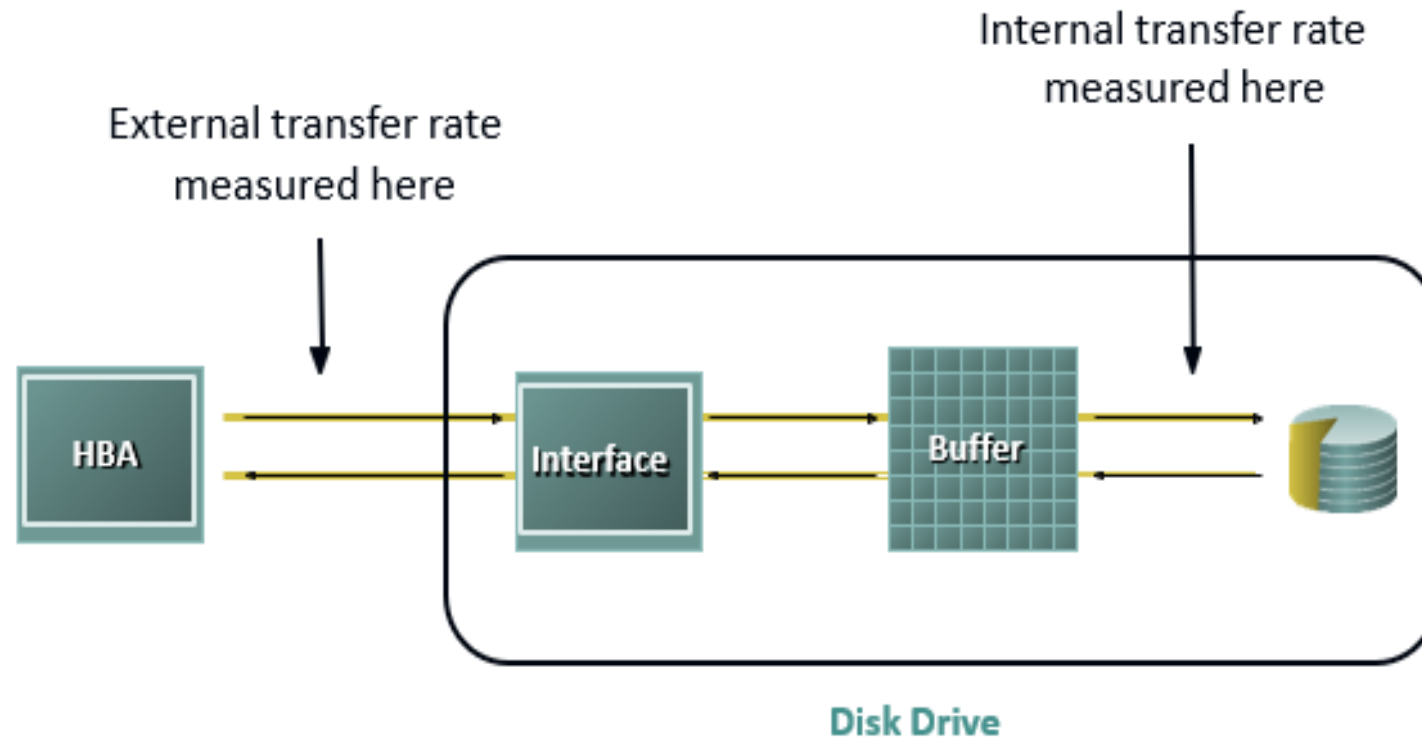
- It is the time taken to position the data under the read/write head by rotating and repositioning the platter.
- Rotational latency is the **time spent waiting** for that required sector to rotate into the head's position.



Disk Drive Performance – 1

Data Transfer Rate

Data transfer rate is also known as transfer rate, and it refers to the average amount of data per unit time that can be delivered by the drive to the Host Bus Adapter (HBA).



Disk Drive Performance – 1

Data Transfer Rate

The Data Transfer Rate describes the MB/second that the drive can deliver data to the HBA. It can be categorized as:

Internal transfer rate - the speed of moving data from the disk surface to the R/W heads on a single track of one surface of the disk.

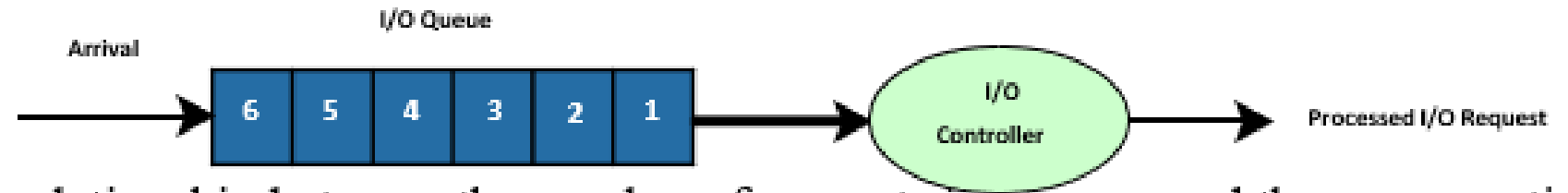
External transfer rate - the rate at which data can be moved through the interface to the HBA.

Disk Drive Performance – 1

Other Factors affecting Disk Drive Performance

Rotation Per Minute (RPM) of a disk drive	Size and number of Platter surfaces	Actuator characteristics
Spindle motor speed	Data recording & encoding factors	Disk response time
File system used	CPU speed	Disk Service Time

Fundamental Laws of Governing Disk Performance



■ Little's Law

- Describes the relationship between the number of requests in a queue and the response time.
- $N = a \times R$
 - “N” is the total number of requests in the queue system
 - “a” is the arrival rate of I/O requests
 - “R” is the average response time

■ Utilisation law

- Defines the I/O controller utilisation
- $U = a \times R_s$
 - “U” is the I/O controller utilisation
 - “R_s” is the *service time*

Calculating the Disk Capacity

Western Digital 26400

Drive Parameters: 13328 cyl • 15 heads • 63 spt • 6448.6 MB

- » 13328 cylinders
- » 15 heads
- » 63 sectors/track
- » 512 B/sector
- » Capacity =
 $13328 \times 15 \times 63 \times 512 =$
6,448,619,520 B



Logical Components of the Host

Application

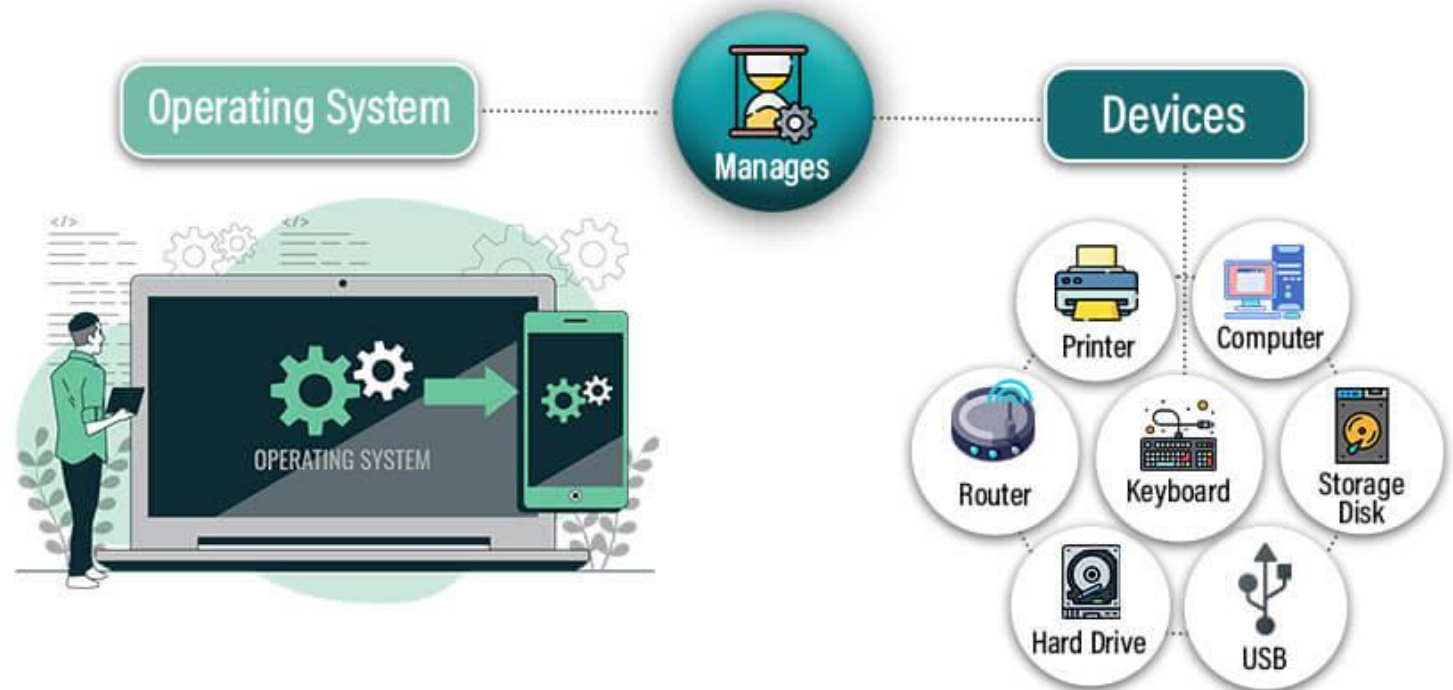
- The interface between a user and the host
- Three-tiered architecture
 - Application UI, Computing Logic and Underlying databases.
- Application data access can be classified as:
 - Block-level access: Data stored and retrieved in blocks, specifying the LBA.
 - File-level access: Data stored and retrieved by specifying the name and path of files.

Logical Components of the Host

Operating System

- Resides between the applications and the hardware.
- Controls the environment.

Device Management in Operating System



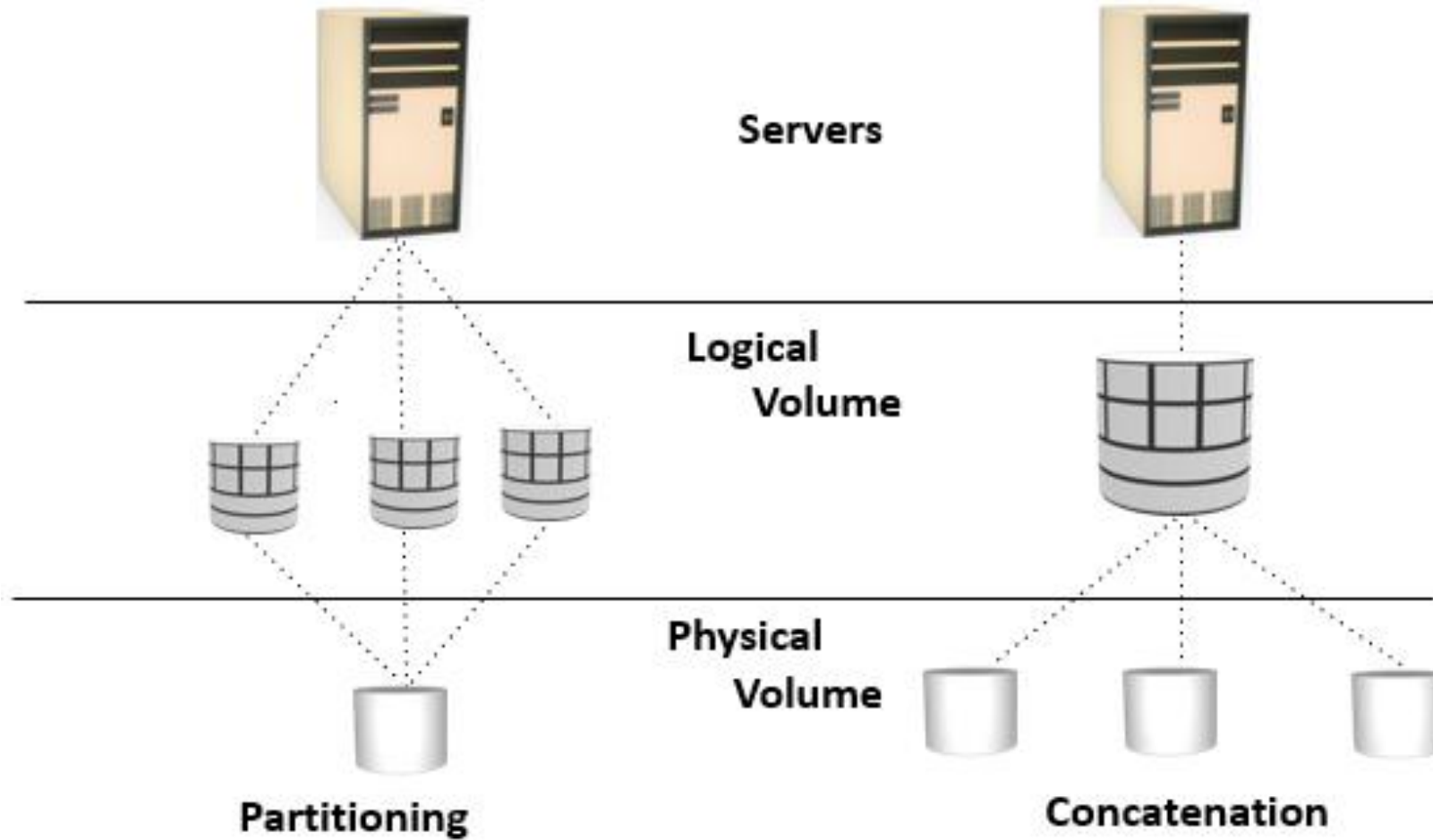
Logical Components of the Host

LVM (Logical Volume Manager)

- Responsible for creating and controlling host level logical storage
 - A physical view of storage is converted to a logical view by mapping
 - Logical data blocks are mapped to physical data blocks
- Usually offered as part of the operating system or a third-party host software
- LVM Components:
 - Physical Volumes
 - Volume Groups
 - Logical Volumes



Logical Components of the Host

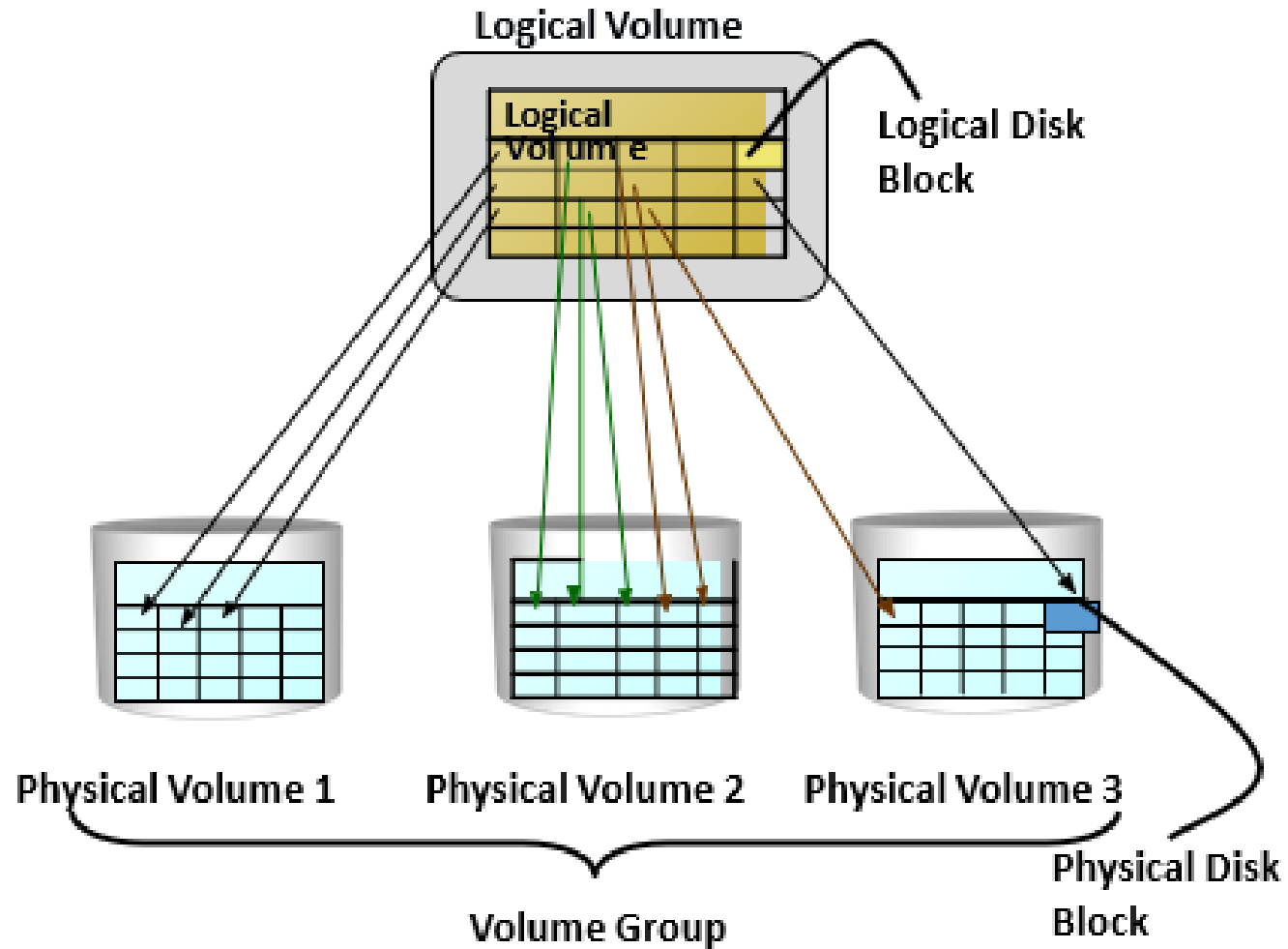


Logical Components of the Host

Volume Group

- One or more Physical Volumes form a Volume Group.
- LVM manages Volume Groups as a single entity.
- Physical Volumes can be added and removed from a Volume Group as necessary.
- Physical Volumes are typically divided into contiguous equal-sized disk blocks.

Logical Components of the Host



Logical Components of the Host

Device Drivers

- Enables the Operating System to recognize the device.
- Provides API to access and control devices.
- They are hardware dependent and Operating System specific.

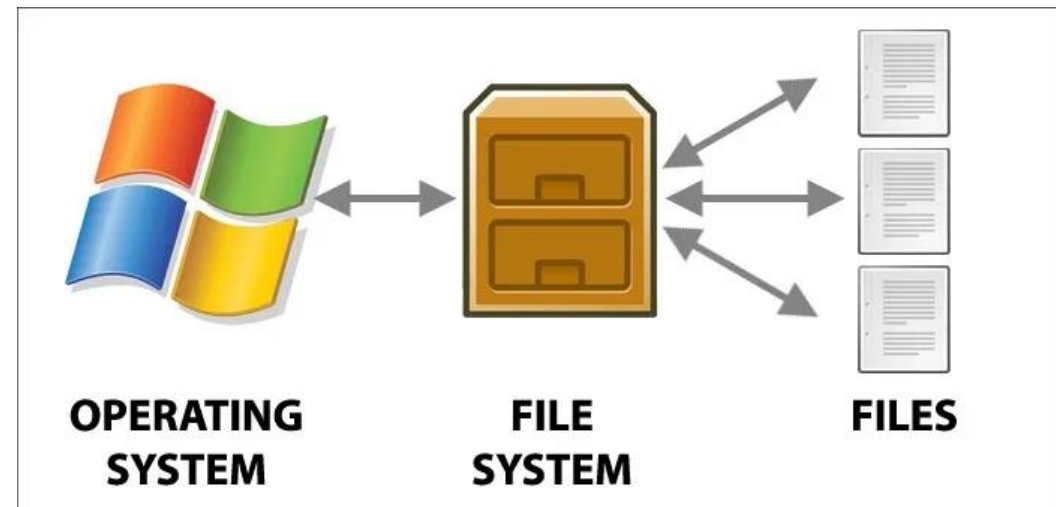
File System

- The file is a collection of related records or data stored as a unit.
- The file system is the organisation of files on disk.

Examples, FAT 32, NTFS, UNIX FS and EXT2/3

File System

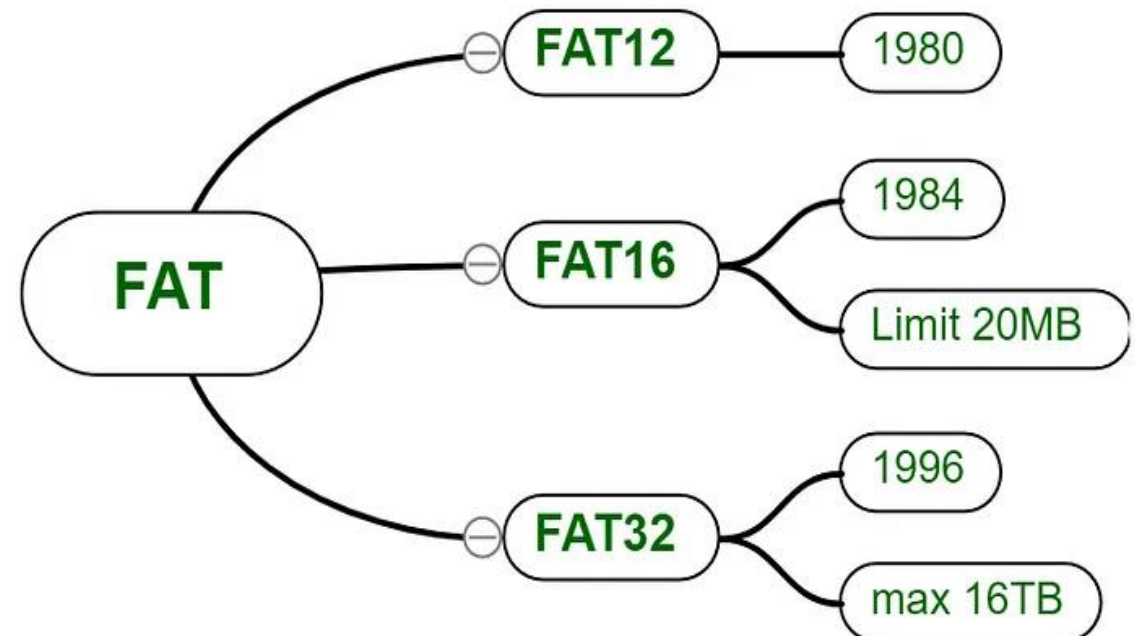
- A file system is a structure used by an operating system to organize and manage files on a storage device such as a hard drive, solid state drive (SSD), or USB flash drive
- File systems are a set of rules and structures used to organize and access data on a computer. A file system provides a directory structure that defines all the files and directories on the computer. It also determines how files are stored and how they are accessed



File System

File Allocation Table (FAT)

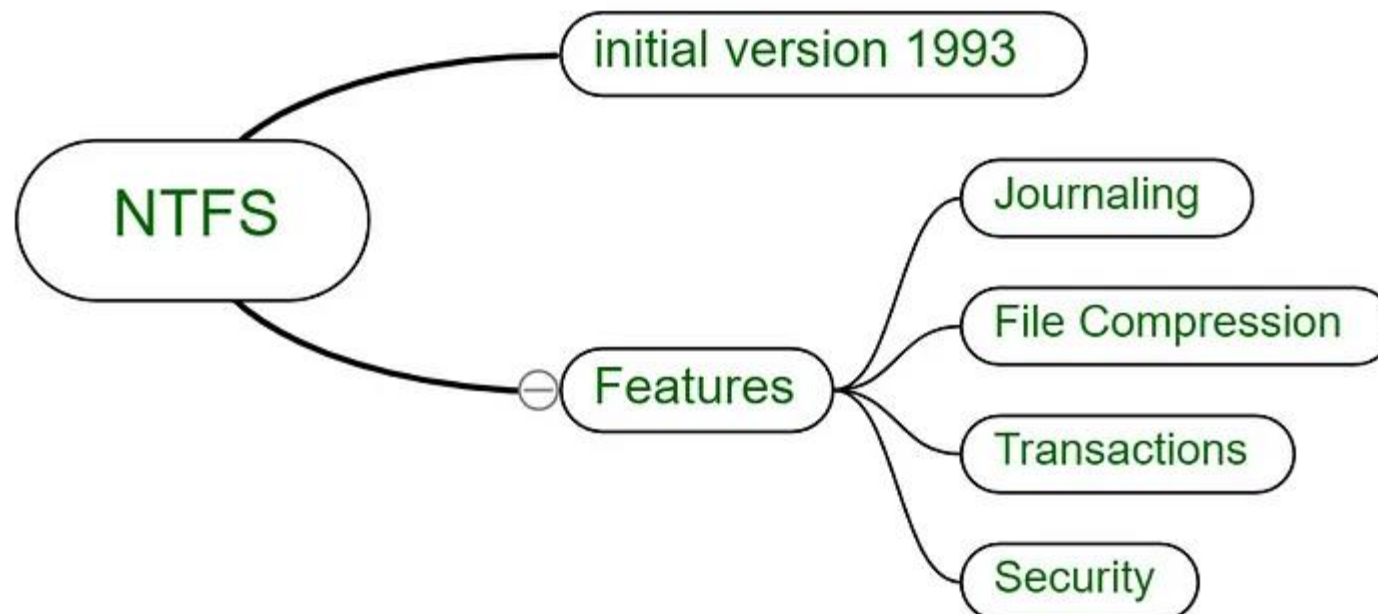
- FAT is one of the oldest and most common file systems. Due to its simple and efficient structure, it is widely used in small portable storage devices.
- Compatibility with different operating systems– FAT is a widely used file system that is compatible with many different operating systems, including Windows, Mac OS, and Linux. This makes it easy to share files between different computers and devices



File System

New Technology File System (NTFS)

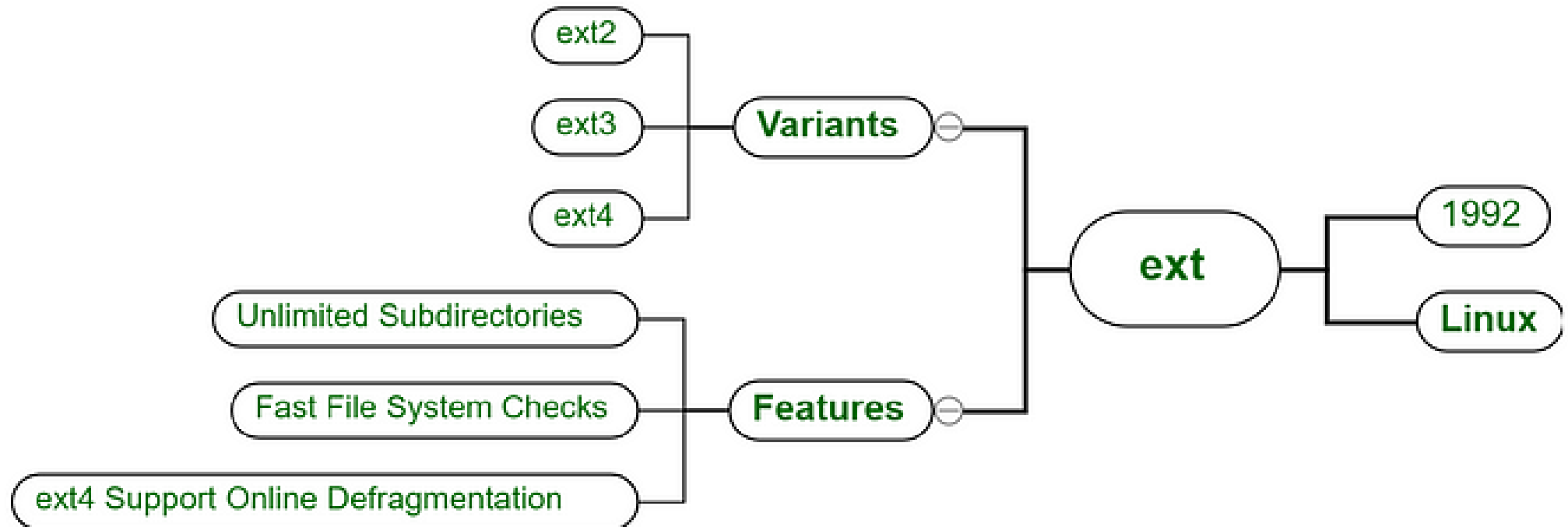
- NTFS is a more advanced file system that was introduced with Windows NT. It supports larger file sizes, better security features, and improved performance compared to FAT. NTFS uses a master file table to keep track of files and directories, and it includes features such as file compression, encryption, and disk quotas. NTFS is the default file system for most modern versions of Windows.



File System

Extension (EXT)

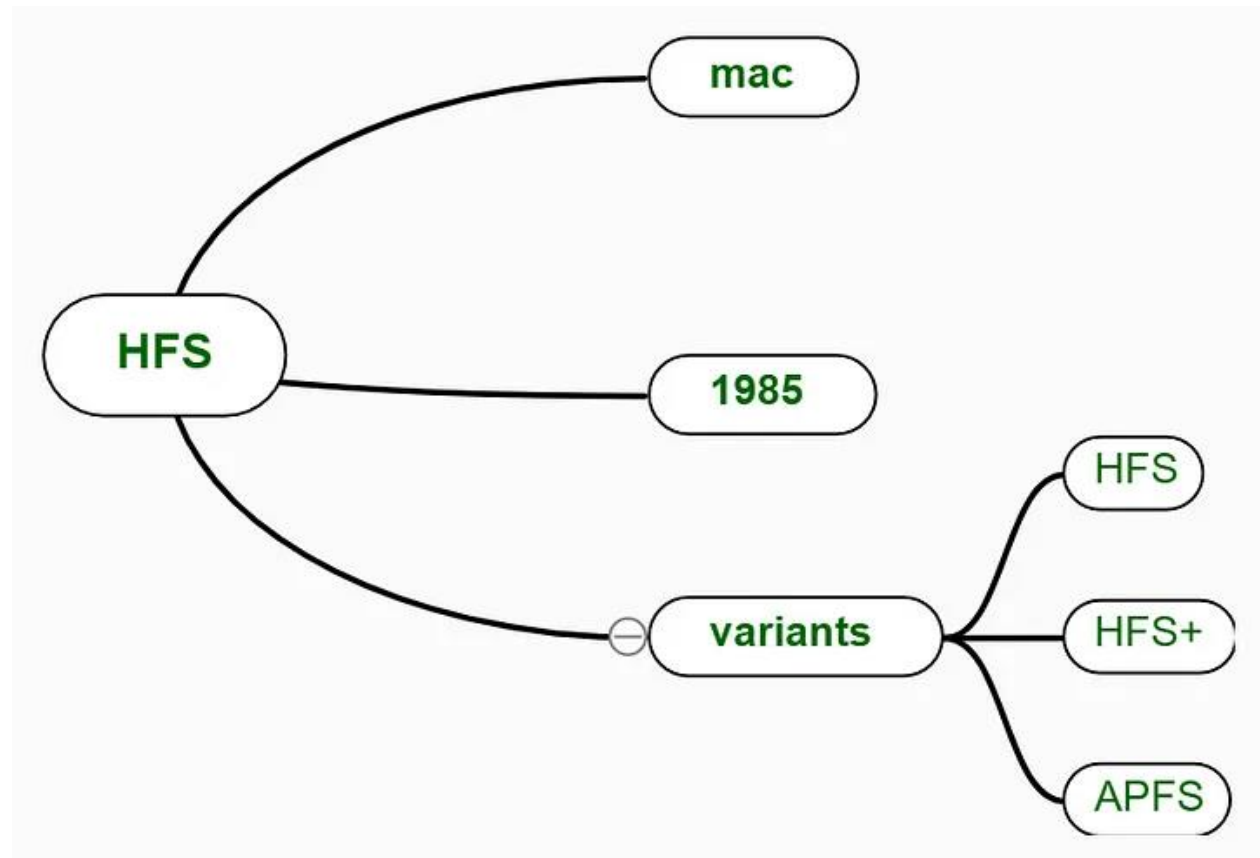
- EXT: It is a file system used in Linux operating systems. It has similar features to NTFS, but it is more efficient



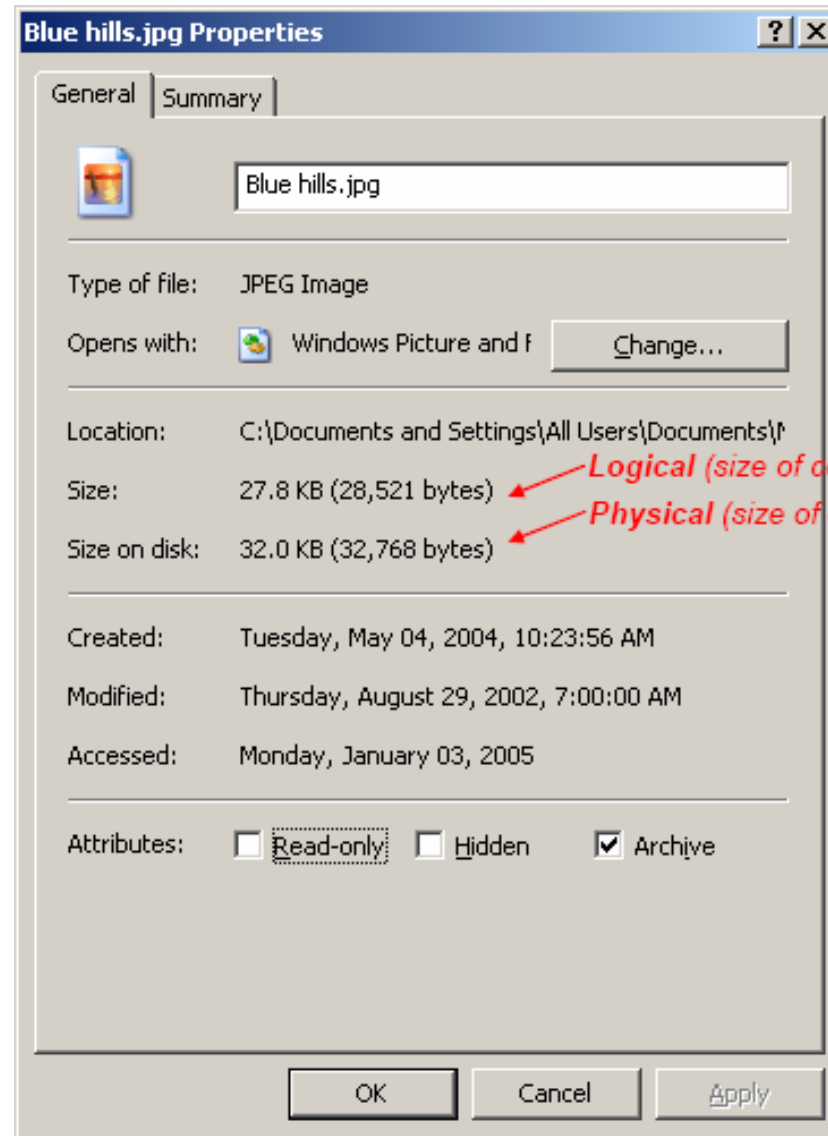
File System

HFS:

HFS (Hierarchical File System) is a file system primarily used in macOS operating systems, organizing files and directories in a hierarchical structure.



Logical vs Physical File Size



Logical (size of contents)

Physical (size of container)

Application Requirements and Disk Performance

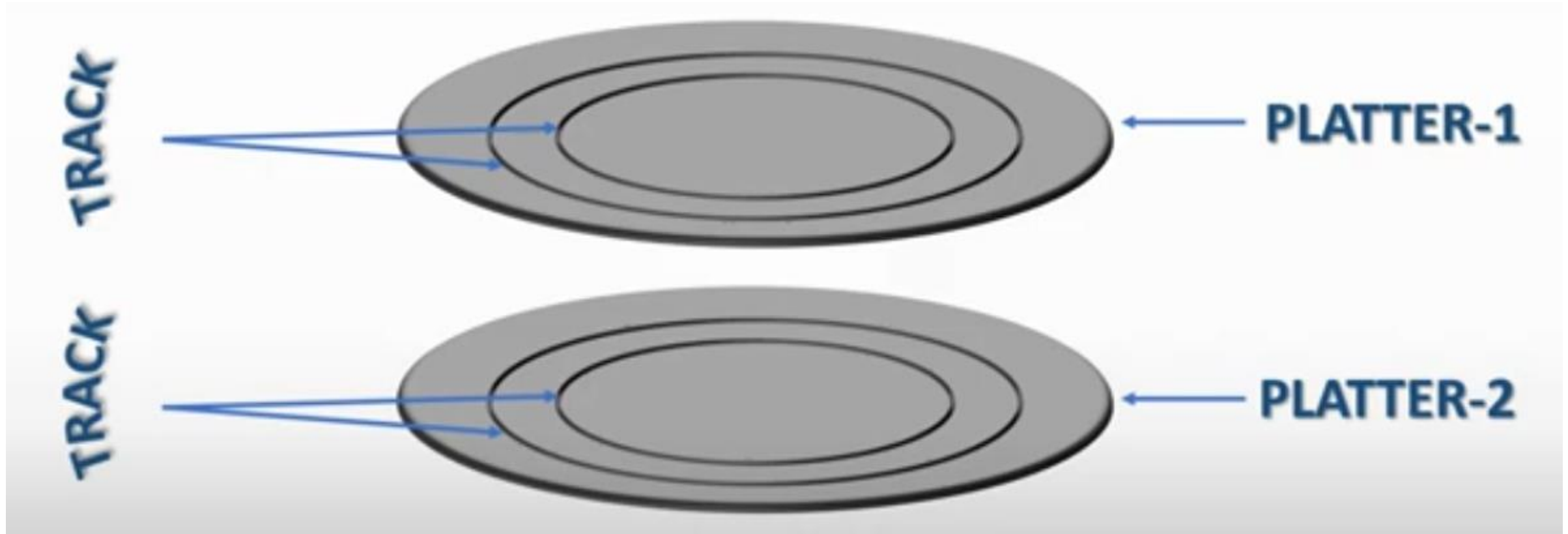
- **Data Throughput:** The amount of data an application needs to read/write per second.
- **Latency:** The time taken to access data. Critical for real-time applications.
- **I/O Operations Per Second (IOPS):** The number of read/write operations an application performs.
- **Data Consistency:** Ensures that data remains accurate and consistent, especially for databases.
- **Scalability:** The ability to handle increasing loads without performance degradation.
- **Access Patterns:** Sequential vs. random access can greatly affect performance.
- **Concurrency:** The number of simultaneous processes accessing the disk.

Recap – Case Study

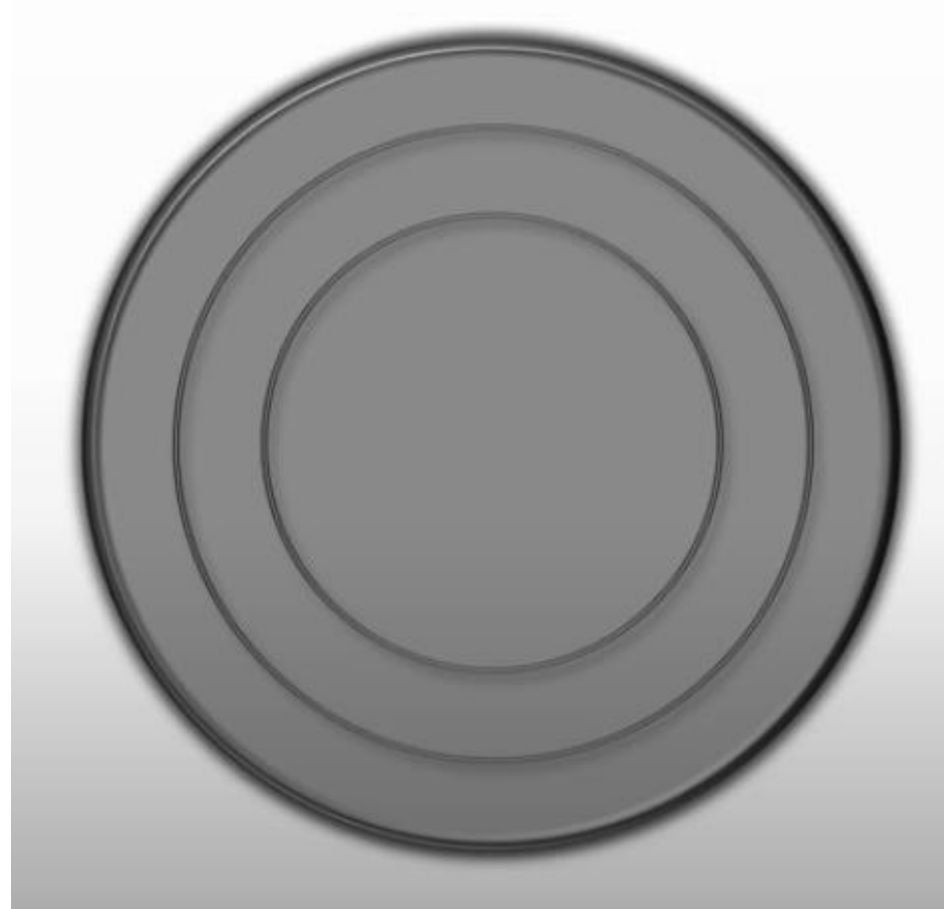
Draw the following diagrams

- Platter
- Cylinder
- Read/Write Head
- Actuator Arm Assembly
- Track
- Sector

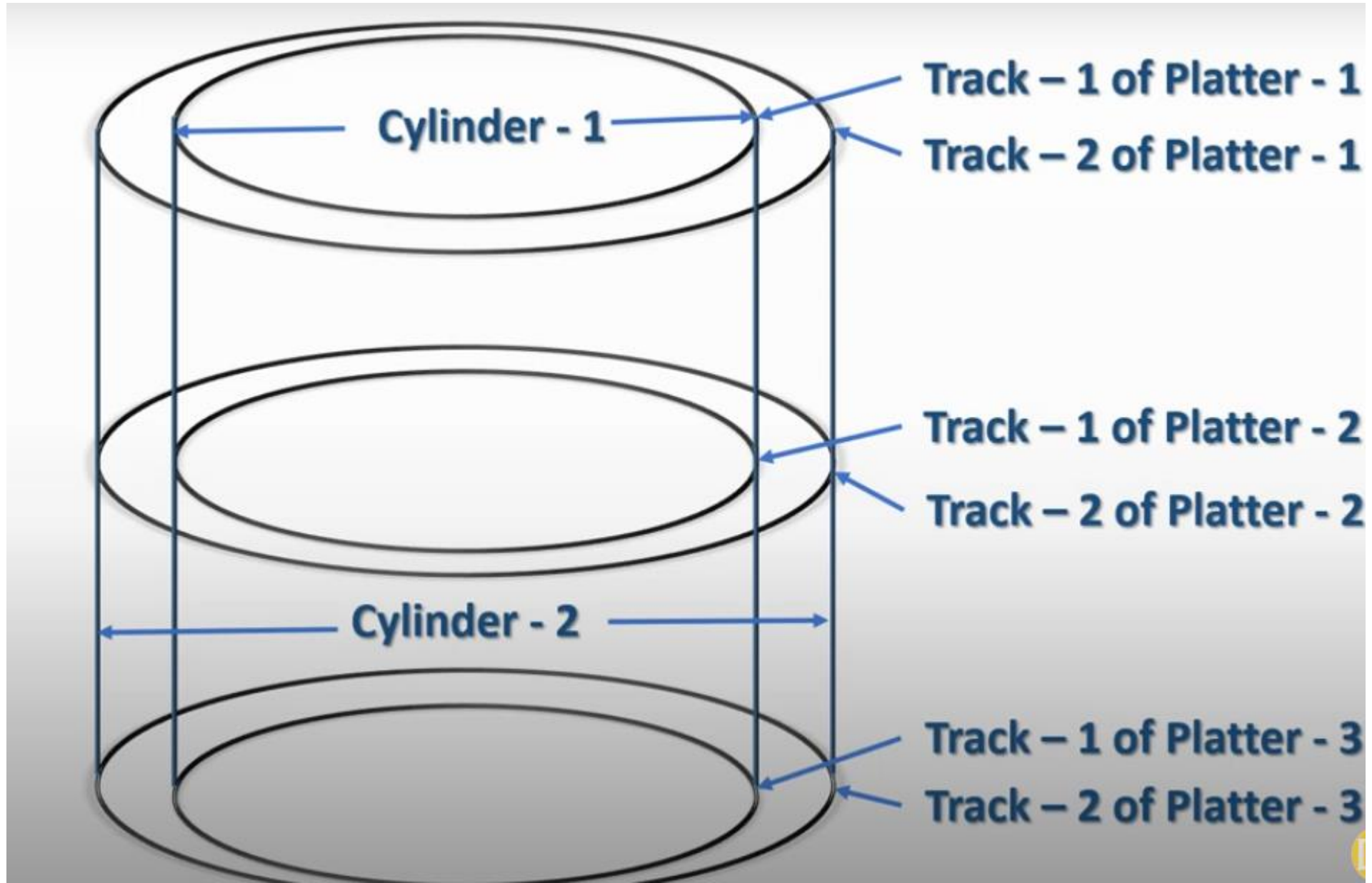
Recap – Case Study



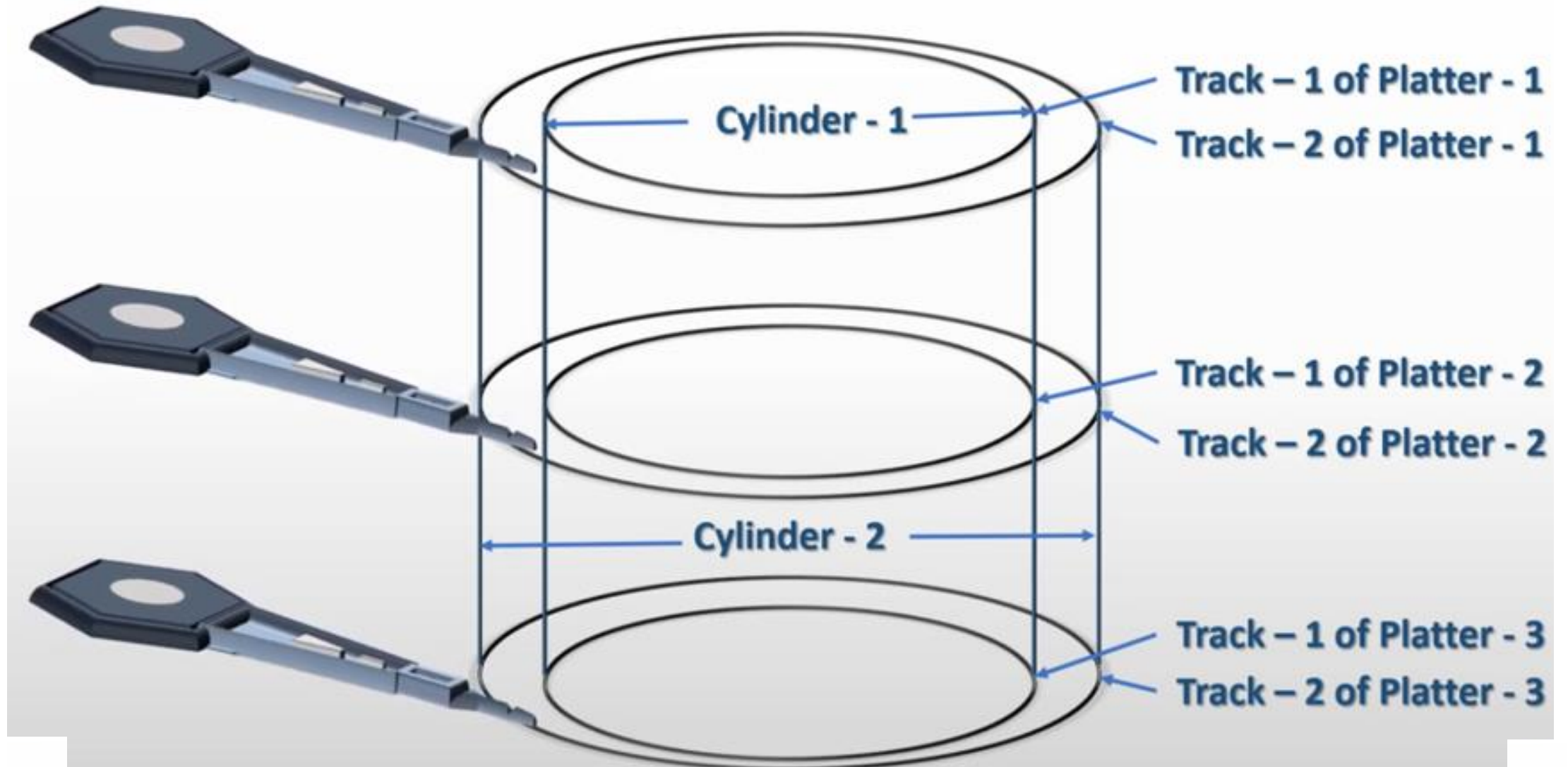
Recap – Case Study



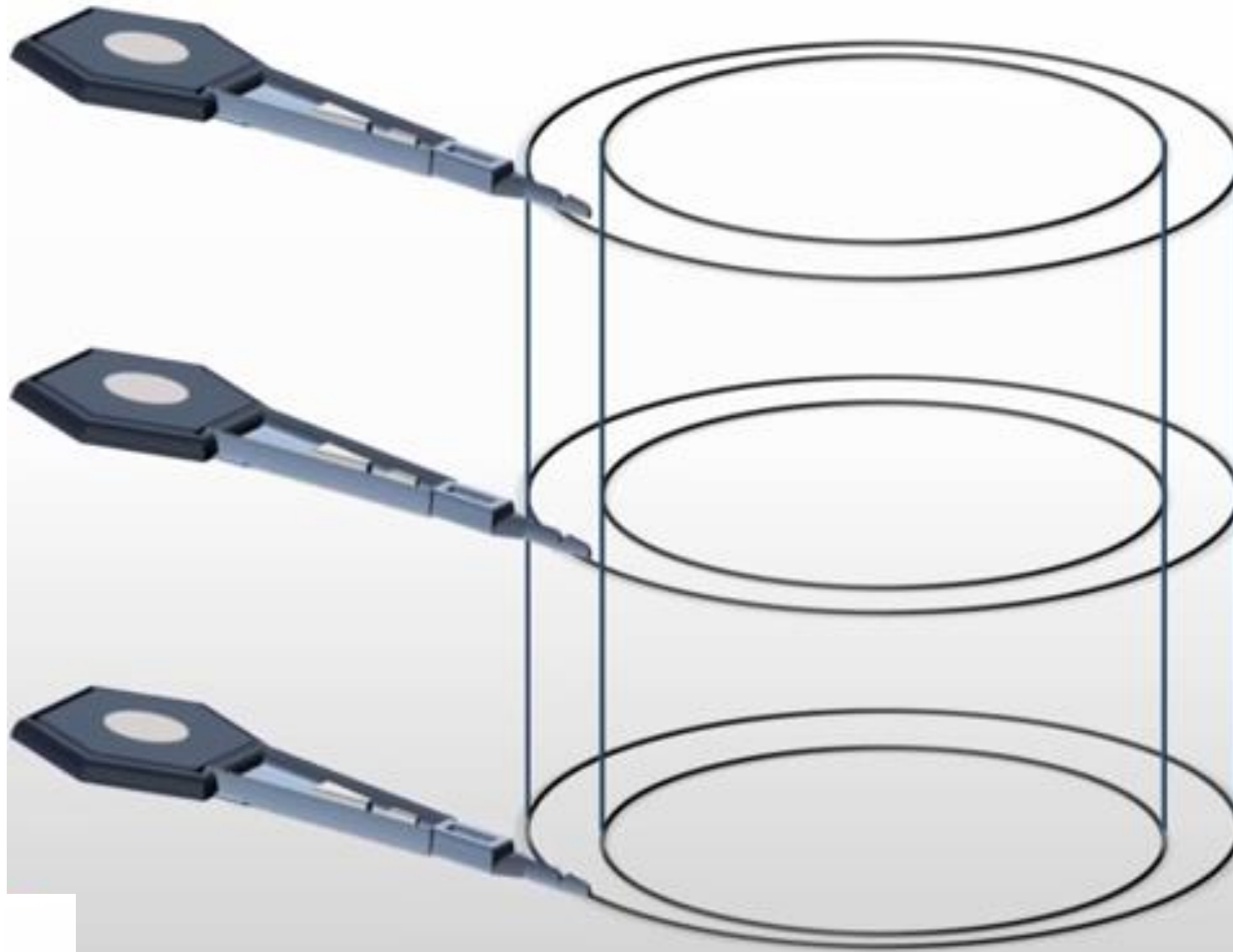
Recap – Case Study



Recap – Case Study

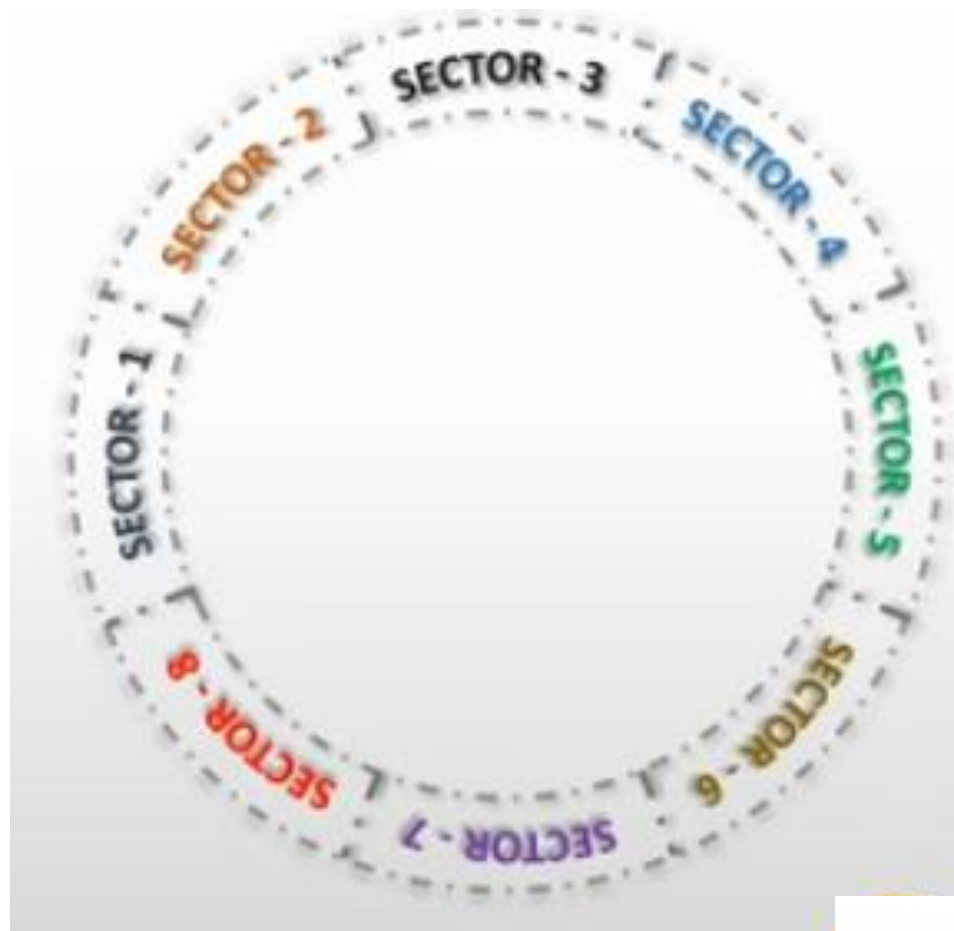


Recap – Case Study



The time required by the read-write head assembly to move / reach / find the desire track / cylinder, i.e. move to one track / cylinder to another, is known as **Seek Time**.

Recap – Case Study



ACCESS TIME

=

SEEK TIME + ROTATIONAL LATENCY

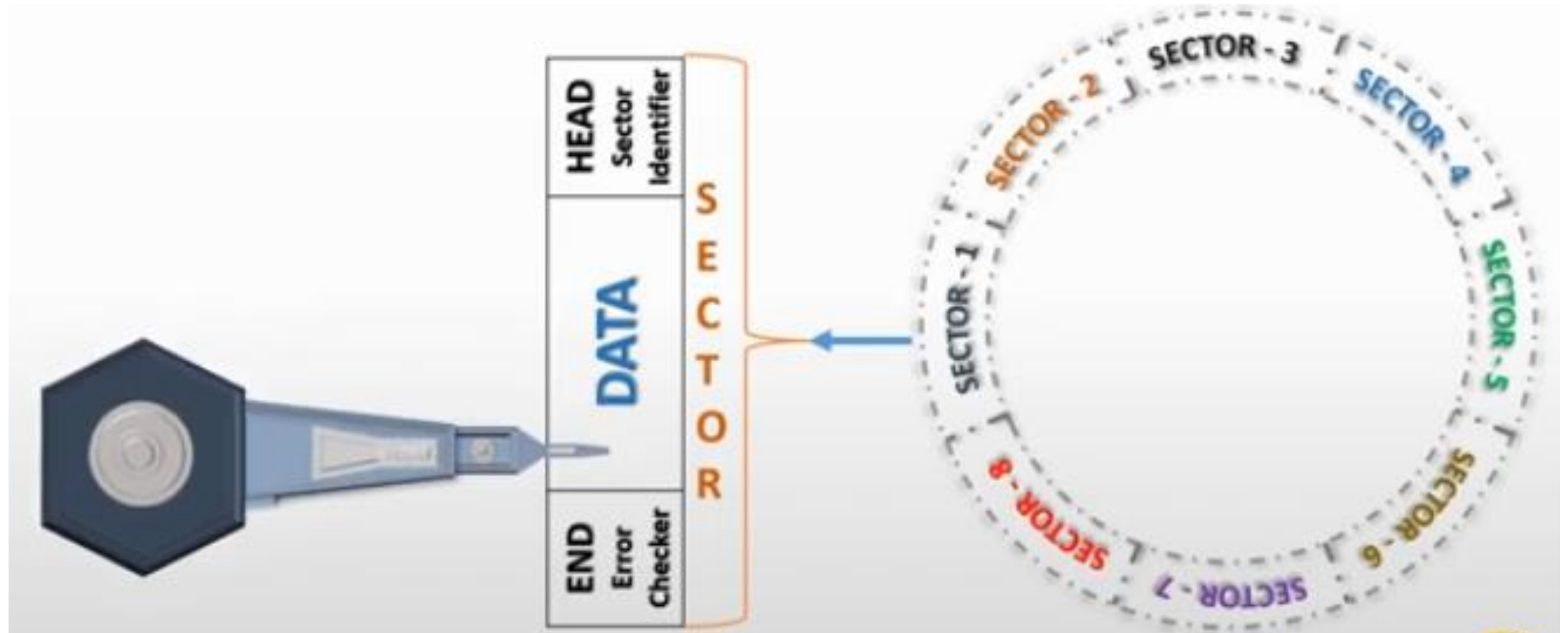


Time taken by the
read-write assembly to
reach desire **Track**.

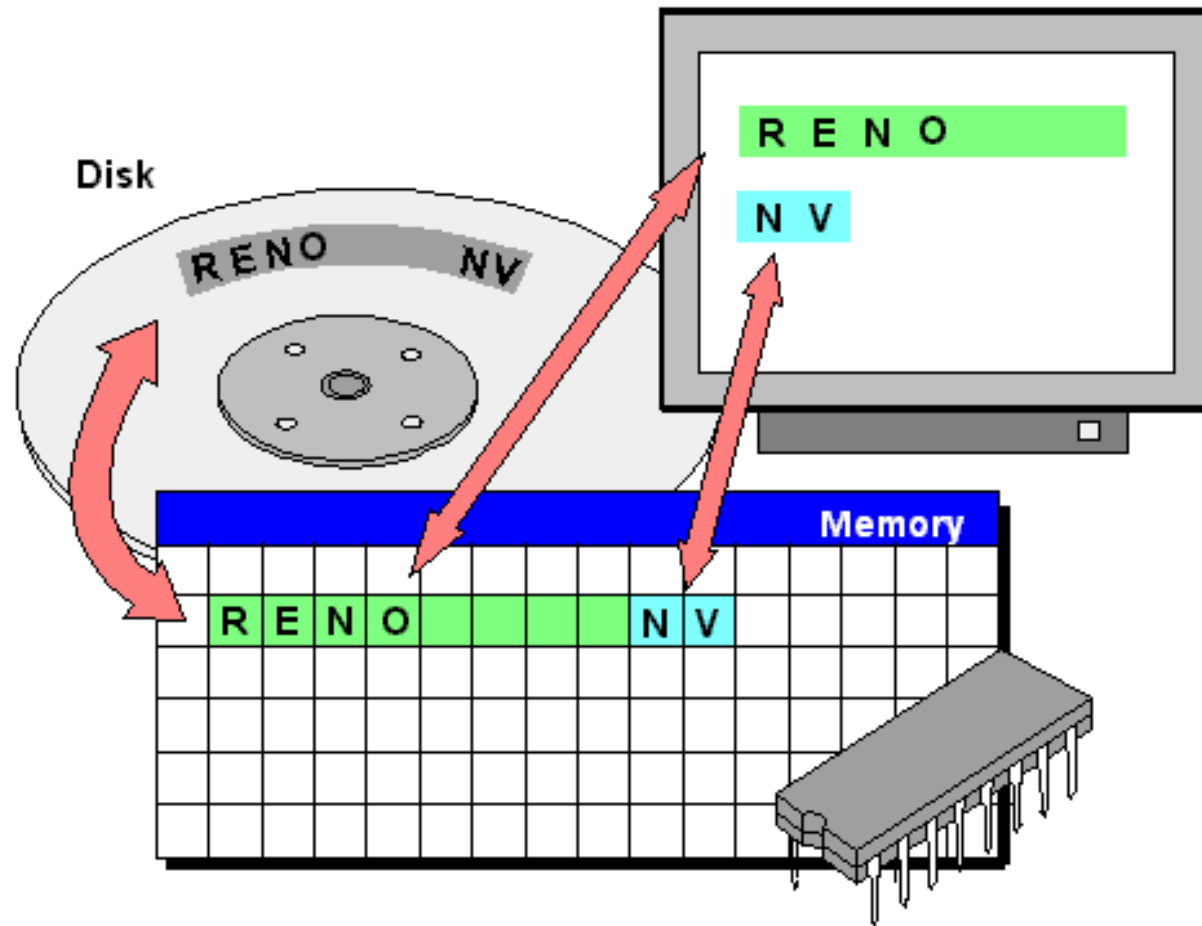


Time taken by the
read-write assembly to
reach desire **Sector**.

Recap – Case Study



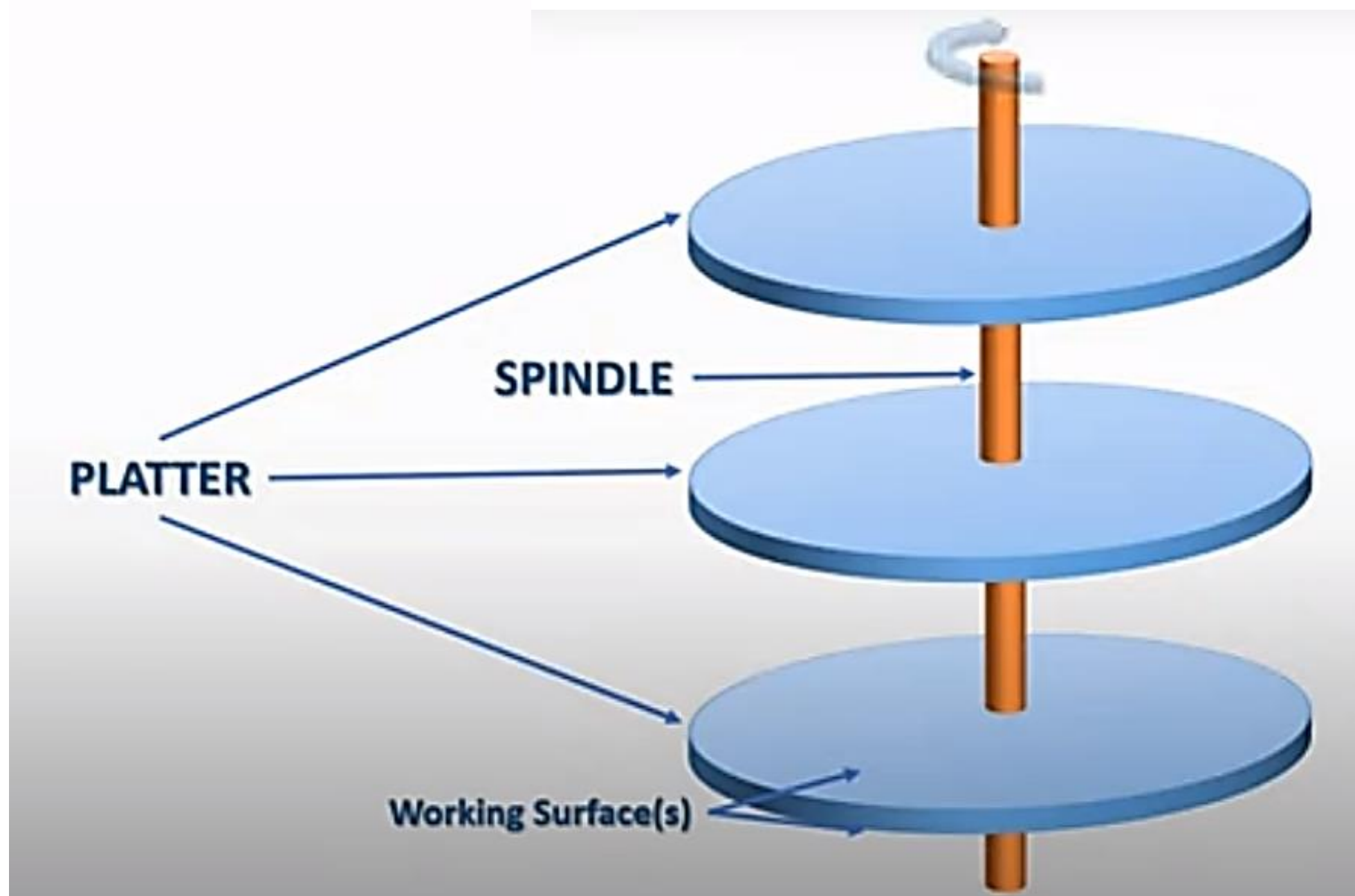
Recap – Case Study



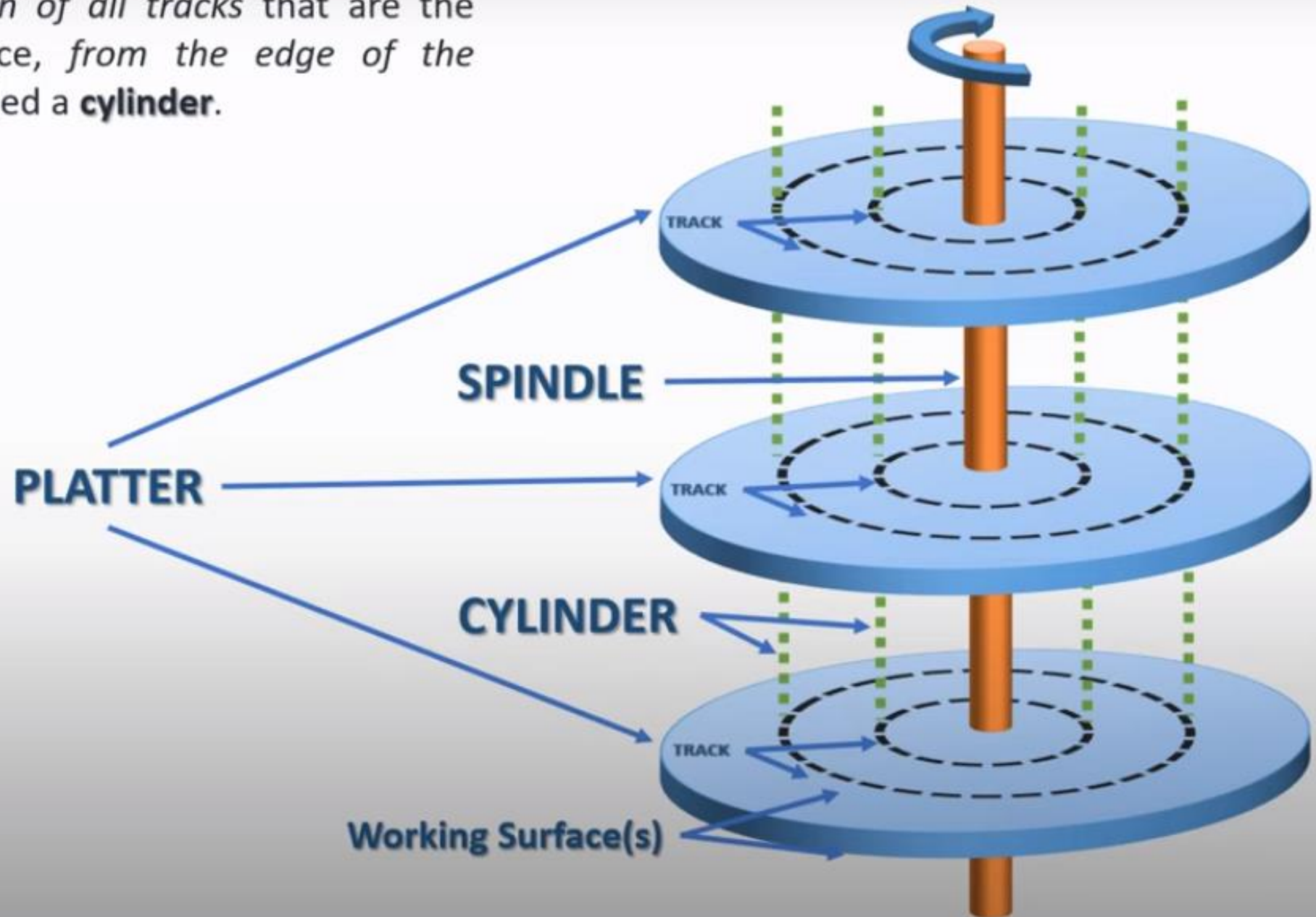
PLATTER

SPINDLE

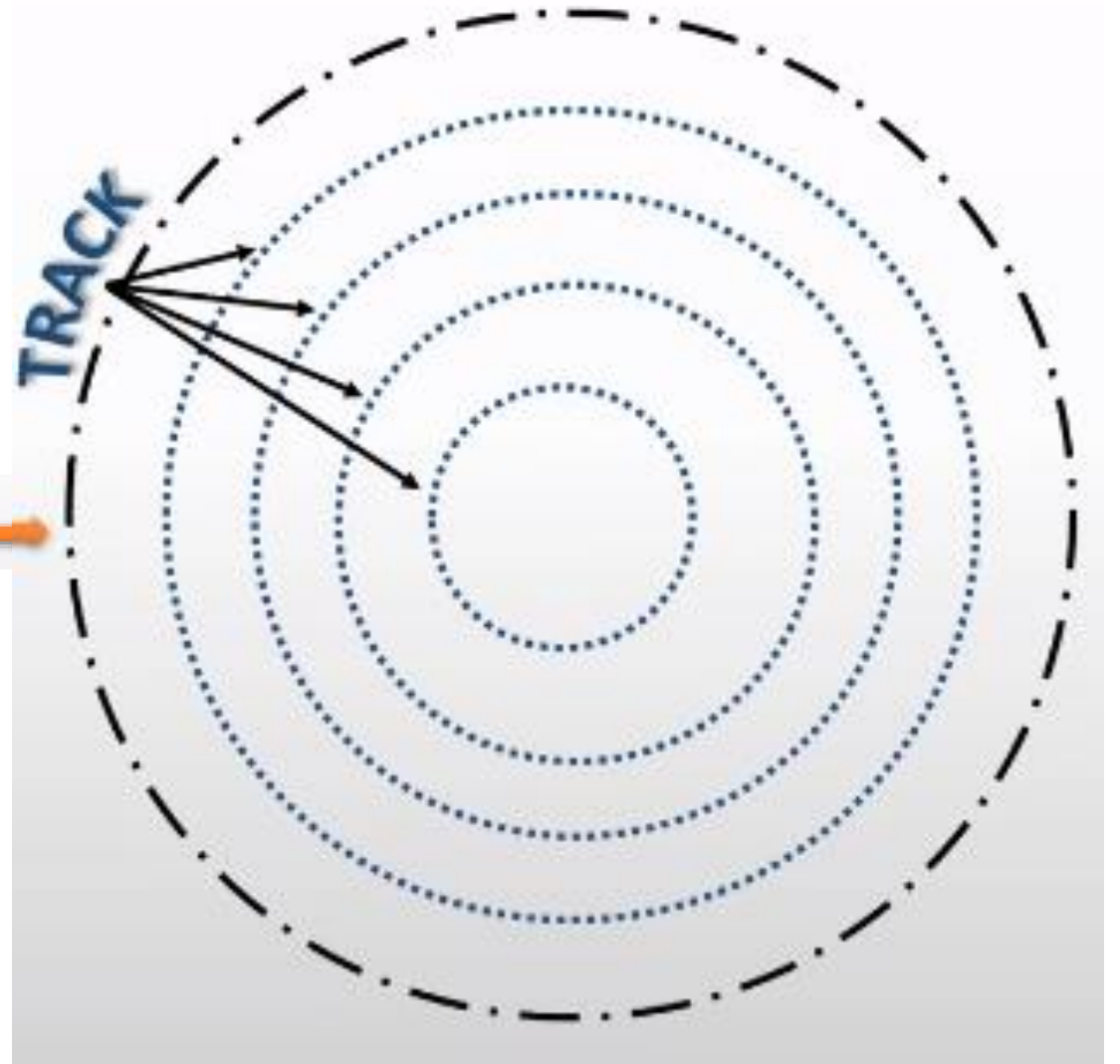
Working Surface(s)



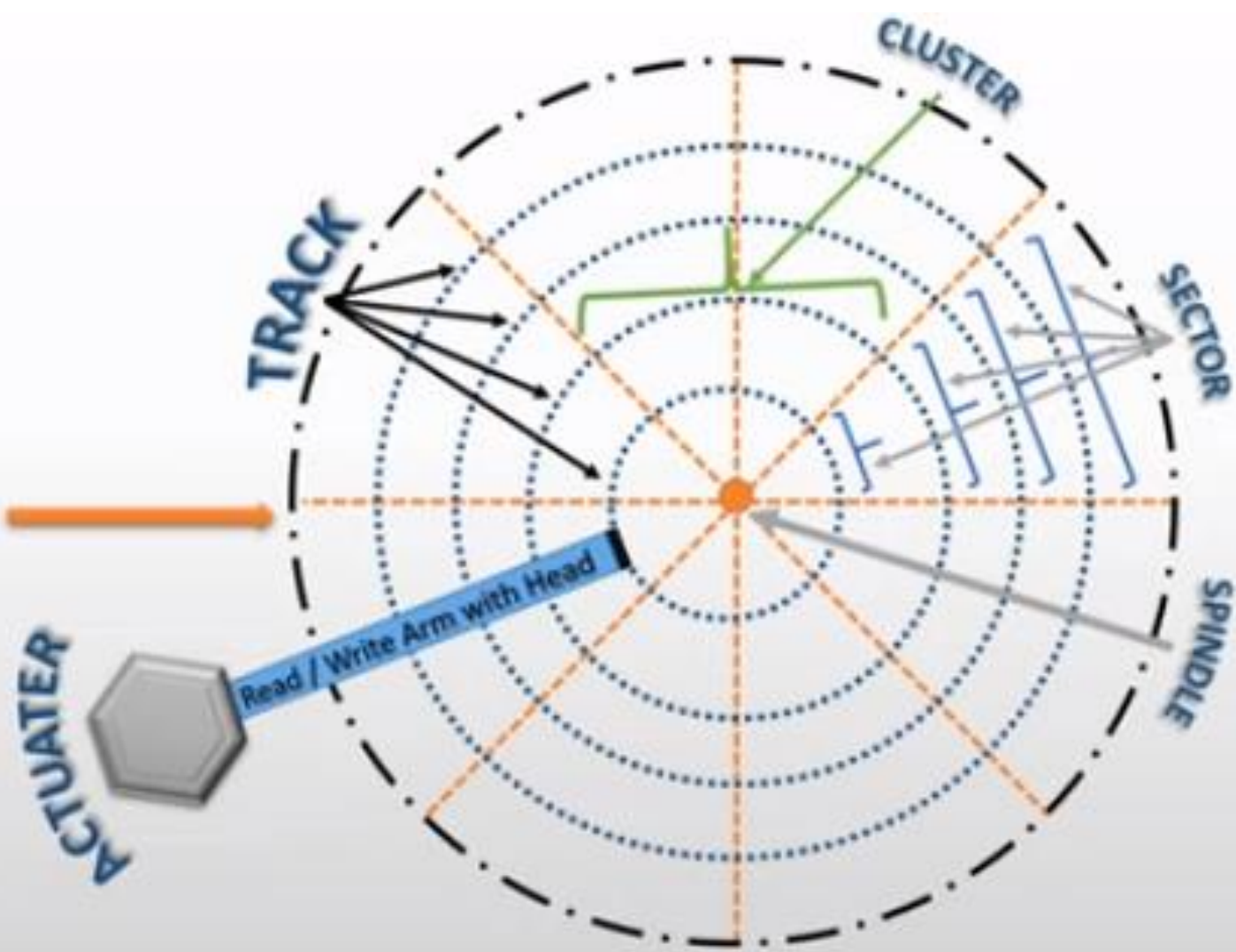
The *collection of all tracks* that are the same distance, from the edge of the platter, is called a **cylinder**.



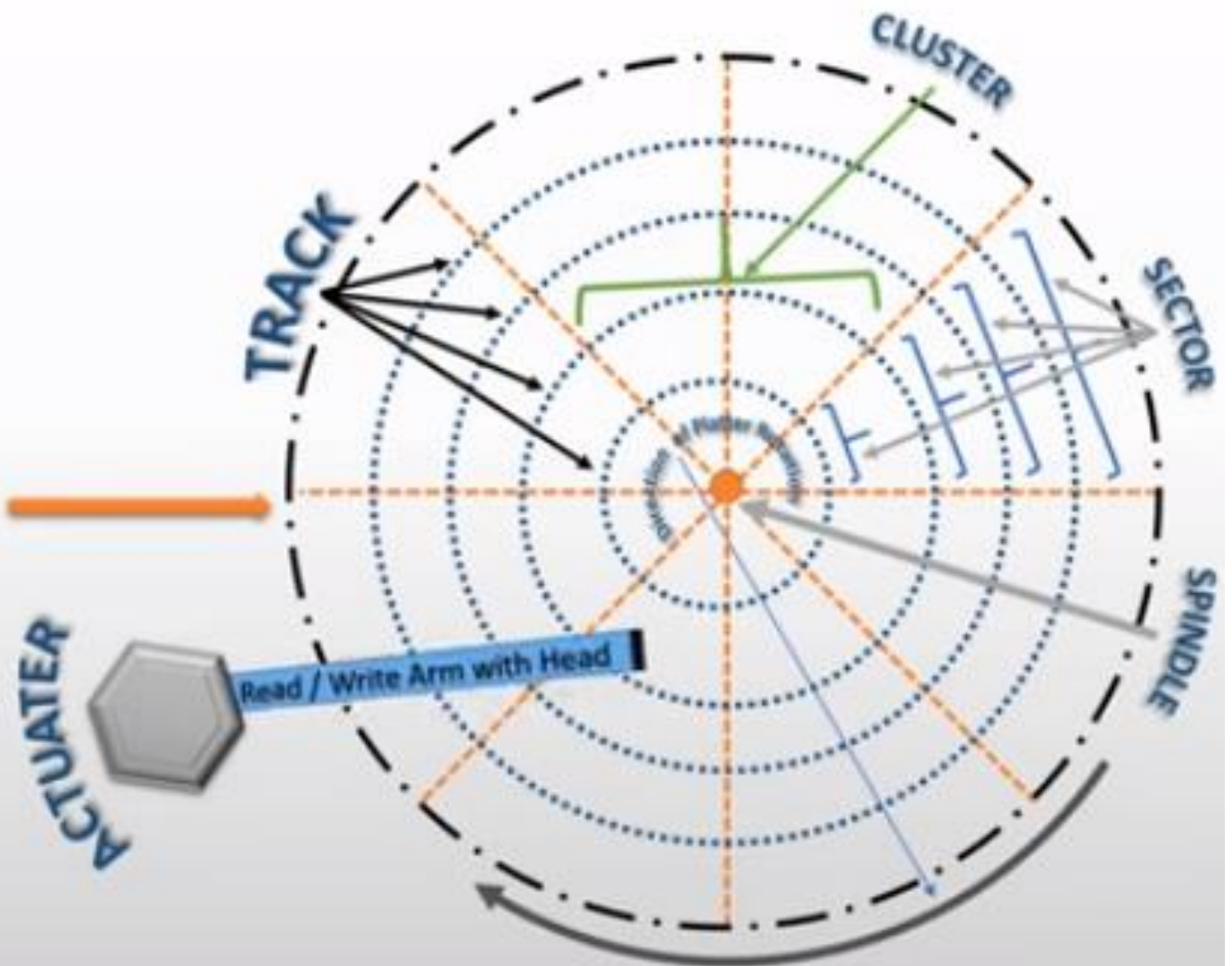
PLATTER



PLATTER



PLATTER



Recap - Capacity and LBA Calculation

1 Sector = 512 Bytes

Capacity = Cylinder x Head x Sector x 512 Bytes

e.g. A hard disk with 1023 Cylinders, 6 Heads and 63 Sectors/Track (CHS=1023/6/63)

- Capacity = $1023 \times 6 \times 63 \times 512 \text{ Bytes} = 386694 \text{ sectors} = 197987328 \text{ B} = 188.8 \text{ MB}$

Suppose a Hard disk with CHS=2040/16/63

To access a sector on Cyl:2000, Head:10, Sect:60
(CHS=2000/10/60)

LBA value = $2000 \times 16 \times 63 + 10 \times 63 + 60$
= 2016690

Summary

- We discussed the Components of a Storage System Environment, Host, Connectivity and Storage.
- Disk Drive Components such as Platter, Spindle, Read/Write Heads, Actuator Arm Assembly, Controller stated.
- Various types of Hard disk connections like SSD & HDD studied
- Server configurations such as Blade server and Rack server mentioned
- Zoned Bit Recording, Logical Block Addressing, Disk Drive Performance and its metrics have been discussed
- Fundamental Laws Governing Disk Performance and Logical Components of the Host have seen.
- Finally the types of File System its Application, Requirements also discussed

Resources

Topics	URL	Notes
Magnetic storage	http://study.com/academy/lesson/magnetic-storage-definition-devices-examples.html	This link explains the magnetic storage technologies with examples
Optical media	http://searchstorage.techtarget.com/definition/optical-media	You will learn in details about optical media with examples and definitions
Solid state storage	http://searchstorage.techtarget.com/definition/solid-state-storage	This link explains about fast solid state storage technology
Components of CPU	https://www.webopedia.com/TERM/C/CPU.html	This link explains about the components of CPU
Tape drive	https://en.wikipedia.org/wiki/Tape_drive	This link gives the details of the magnetic tape drive features

Resources

Topics	URL	Notes
Storage System	https://searchstorage.techtarget.com/tip/The-components-that-make-up-a-well-balanced-data-storage-system	The link explains in details all aspects of a balanced data storage system
Magnetic storage	http://study.com/academy/lesson/magnetic-storage-definition-devices-examples.html	This link explains Magnetic Storage: Definition, Devices and Examples
Optical media	http://searchstorage.techtarget.com/definition/optical-media	This link explains about different types of Optical media (CD, DVD)
Solid State Devices	http://searchstorage.techtarget.com/definition/solid-state-storage	This link explains about Solid State Storage Technology
Performance assessment of Disk Drive	https://www.pctechguide.com/hard-disks/hard-disk-hard-drive-performance-transfer-rates-latency-and-seek-times	This link explains various parameters of Drive Performance - transfer rates, latency and seek times
Hard disk drive	https://www.explainthatstuff.com/harddrive.html	This link explains about hard disk drive and its components. It presents details about structure of a hard disk drive and its functioning

Resources

Topics	URL	Notes
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Types of storage media

<https://www.youtube.com/watch?v=NgVm2kUmrYo>

This video explains and compares different types of storage media

Storage system environment

<https://www.youtube.com/watch?v=-8hAOsQnbK8>

This video explains storage system environment

Topics	URL	Notes
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Hard Disk Drive

<https://www.explainthatstuff.com/harddrive.html>

This link explains the hard disk drive and its components. It presents details about structure of a hard disk drive and its functioning

Topics	URL	Notes
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Hard Disk Drive in motion

<https://www.youtube.com/watch?v=p-JJp-oLx58>

This video discusses the components and working of a hard disk drive. It shows how the components work when the hard disk is in motion

Resources

Topics	URL	Notes
Storage Media	https://www.youtube.com/watch?v=NgVm2kUmrYo	This video explains various types of storage media
HDD working	https://www.youtube.com/watch?v=p-JJp-oLx58	This video illustrates various components of hard disk drive and shows their working
Storage Area Network	https://www.youtube.com/watch?v=7et3bmXhr8w	This video explains the building blocks of SAN
Introduction to NAS	https://www.youtube.com/watch?v=KXQUpJWTrIA	This video covers various aspects of NAS in details.

Topics	URL	Page Number
Storage Systems Architecture	http://read.pudn.com/downloads168/ebook/772307/STF2-0SectionIntroduction.pdf	1 to 7
Information Storage and Management	https://www.pdfdrive.com/information-storage-and-management-volume-1-of-rnhtnet-d17363479.html	1-317

References

- EMC Education Services, “Information Storage and Management: Storing, Managing, and Protecting Digital Information”, Wiley Publishing Inc., 2nd edition, 2012.
- <https://www.slideserve.com/ansel/storage-system-environment>
- <https://networkinterview.com/blade-server-vs-rack-server-vs-tower-server/>
- <https://www.sunbirdcim.com/glossary/blade-server>
- <https://medium.com/@eyupilis/file-systems-and-their-role-in-operating-systems-158cbf782530>
- <https://www.kingston.com/en/blog/personal-storage/understanding-file-systems>
- <https://benkixmiller.com/2023/10/08/chs-vs-lba-addressing/>
- <http://www.c-jump.com/CIS24/Slides/DiskDrives/DiskDrives.html>





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